ESSENTIAL MATHEMATICS GRADE 10: COMMERCIAL ARITHMETIC 1

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.8 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Essential Mathematics
Strand	Strand 2.0: Measurements and Geometry
Sub-Strand	Sub-Strand 2.8: Commercial Arithmetic 1
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Budget: preparing and using a budget for clubs or societies in a school setting – Discounts: calculating discounts and percentage discounts in different shopping situations – Commission: determining commission and percentage commission paid to salespersons for sale of goods and services – Profit and Loss: determining percentage profit and loss made in the sale of goods – Currency Conversion: converting currencies using exchange rate tables (buying and selling rates, Kenya Shillings to other currencies and vice versa)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) prepare a budget for use in clubs or societies; b) calculate discount and percentage discounts in different situations; c) determine commission and percentage commissions in different situations; d) determine percentage profit and loss made in the sale of goods; e) convert currencies using exchange rates tables; f) recognise the role of commercial arithmetic in day-to-day life.
Core Competencies	– Citizenship: The learner develops ethical and digital citizenship skills when obtaining digital or printed currency exchange rates charts/tables and working out currency conversion from Kenya Shillings to any other currencies. – Critical Thinking and Problem Solving: The learner acquires interpretation and inference skills when role-playing shopping scenarios where discounts are given and working out the discounts and the percentage discount.
Core Values	– Peace: The learner respects self and others by listening to others' opinions while brainstorming scenarios under which commissions are paid to salespersons for sale of goods and services. – Patriotism: The learner displays loyalty by showing love for one's own country's currency when obtaining digital or printed currency exchange rates tables and working out currency conversion from Kenya Shillings to any other currencies.
Science & Engineering Practices	– Asking Questions and Defining Problems: Learners pose questions about why prices differ at different market stalls in Nairobi's Gikomba or Mombasa's Marikiti market, and define the financial problem of budgeting limited club funds. – Analysing and Interpreting Data: Learners extract and interpret data from currency exchange rate tables and printed price lists to compute discounts, commissions, profit, and loss. – Constructing Explanations: Learners construct mathematical explanations for why a Nakumatt/Naivas promotion discount

	benefits the buyer and how a salesperson's commission incentivises sales. – Engaging in Argument from Evidence: Learners use calculated percentage profit/loss figures to argue whether a small-scale maize trader at Wakulima Market made a sound business decision.
Pertinent & Contemporary Issues (PCIs)	– Financial Literacy: The learner's financial skills are enhanced when obtaining digital or printed currency exchange rates charts/tables and working out currency conversion from Kenya Shillings to any other currencies, and when preparing budgets for school clubs.
Career Connections	– Sole Proprietorship / Small Business Owner: Understanding profit, loss, discounts, and budgeting is directly applicable to running a small business such as a kibanda food stall or a mitumba clothing stall. – Sales Agent / Commission-Based Salesperson: Computing commission and percentage commission is the core skill required for careers in insurance sales, real estate agency, and mobile-money agent operations across Kenya. – Accountant / Bookkeeper: Budget preparation and profit-and-loss determination underpin entry-level bookkeeping and accounting roles in Kenyan SMEs and cooperative societies (saccos). – Tourism and Hospitality Industry: Currency conversion skills are essential for front-office staff at hotels and tour operators in Nairobi, Mombasa, and Diani dealing with international visitors. – Journalism / Media: Financial literacy including reading exchange-rate tables and interpreting economic data is needed by business journalists and media analysts.
Focus for Lessons	This unit builds learners' financial literacy by anchoring all five content topics — budgeting, discounts, commission, profit/loss, and currency conversion — in the lived economic realities of Kenyan markets, school clubs, and cross-border trade. Learners move from preparing a simple club budget to role-playing market transactions and finally to reading real (or simulated) Central Bank of Kenya exchange-rate tables, progressively building the commercial arithmetic skills they need for informed personal and professional financial decision-making.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can Amina — and every Kenyan — use commercial arithmetic to make smart, fair, and informed money decisions in markets, clubs, and across borders? KICD KEY INQUIRY QUESTIONS: 1. Why do we need a budget? 2. Why are discounts and commissions necessary in trade?
Anchoring Phenomenon	A Form 4 student named Amina from Kisumu has saved KSh 4,500 from selling mandazi at school and wants to use the money in three ways: contribute to her school Environmental Club's end-of-term trip to Lake Victoria, buy a second-hand smartphone advertised at "20% off" at a Kondele electronics stall, and send the remaining money to her aunt in Uganda via a mobile-money transfer. When Amina sits down to plan, she quickly realises that the "20% discount" still leaves her short, the mobile-money agent charges a commission, and the Uganda shilling exchange rate she saw yesterday has changed today. Students are puzzled: How can the same KSh 4,500 produce such different outcomes depending on discounts, commissions, exchange rates, and whether a transaction results in a profit or a loss — and how can Amina (or any Kenyan) make smart financial decisions using mathematics?
Storyline Thread	Lesson 1 — Anchoring Phenomenon & Budgeting: Introduce Amina's dilemma with the KSh 4,500. Learners brainstorm how their own school clubs raise and spend money, then draft a simple income-and-expenditure budget for a hypothetical Environmental Club trip to Lake Victoria, identifying fixed and variable costs. Connect budget shortfalls back to the phenomenon. Lesson 2 — Discounts and Percentage Discounts (Part 1): Using Amina's smartphone purchase as context, learners role-play a shopping scenario at a Kondele electronics stall. They calculate discount amounts and percentage discounts from marked prices,

	and explore how "20% off" is computed. Predict phase: estimate the sale price before calculating. Lesson 3 — Discounts and Percentage Discounts (Part 2 / Practice): Learners work through varied Kenyan retail contexts (Naivas Weekend Specials, mitumba clothing, school-uniform shops). They compare discount offers and determine which deal gives better value, reinforcing percentage discount calculations with peer discussion and presentations. Lesson 4 — Commission and Percentage Commission: Introduce the M-PESA agent in Kericho as a supporting phenomenon. Learners brainstorm real scenarios where commissions are paid (insurance agents, real estate, mobile-money, market brokers). They calculate commission amounts and percentage commissions from given transaction values, role-playing salesperson-and-customer exchanges. Lesson 5 — Profit and Loss (Part 1 — Concepts & Calculation): Using the Wakulima Market maize trader as a supporting phenomenon, learners determine cost price, selling price, profit or loss, and percentage profit or loss. They role-play buying-and-selling scenarios involving local goods (githeri, beans, second-hand clothes) and make presentations to peers. Lesson 6 — Profit and Loss (Part 2 — Application & Problem Solving): Learners tackle multi-step profit-and-loss problems set in Kenyan contexts (a small restaurant selling nyama choma, a school tuck-shop selling mandazi). They consolidate percentage profit/loss skills and connect findings back to Amina's mandazi-selling enterprise in the anchoring phenomenon. Lesson 7 — Currency Conversion: Using simulated Central Bank of Kenya forex bureau exchange-rate tables, learners convert Kenya Shillings to Uganda shillings, US dollars, and euros and back again. They explore why buying rates differ from selling rates (the forex bureau's commission), linking this to Lesson 4. Learners solve Amina's cross-border transfer problem. Lesson 8 — Synthesis, Model Revision & Assessment: Learners revisit the full Amina scenario and produce a written financial plan that incorporates a budget, discount calculation, commission deduction, and currency conversion — demonstrating all six learning outcomes. Peers review each other's plans using the KICD assessment rubric criteria, and the class updates the Driving Question Board with final answers.
Supporting Phenomena	– A maize trader at Wakulima Market in Nairobi buys a 90 kg bag of maize for KSh 2,700 and sells it at KSh 28 per kg — students investigate whether she makes a profit or a loss and by what percentage. – A Safaricom M-PESA agent in Kericho earns a commission every time a customer sends money; students observe the agent's commission schedule and calculate her monthly earnings for different transaction volumes. – A Naivas Supermarket "Weekend Special" flyer advertises sukuma wiki at "Buy 3 bunches, get 1 free" — students unpack this as an implied percentage discount and compare it with a rival stall's straightforward 15% price reduction. – A Kenyan student returning from a school exchange visit to Tanzania must convert 50,000 Tanzanian shillings back to Kenya shillings using the day's buying and selling rates from a forex bureau at JKIA — students explore why the buying rate differs from the selling rate and what that difference means.

LESSON 1 (40 minutes): Amina's KSh 4,500 — Anchoring the Phenomenon & Building a Club Budget	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Launch the anchoring phenomenon through Amina's KSh 4,500 dilemma and establish what a budget is, why clubs need one, and how income and expenditure are categorised as fixed or variable costs.
Knowledge	– Define a budget as a plan that shows expected income and expected expenditure over a given period. – Distinguish between fixed costs (costs that do not change regardless of the number of participants, e.g., bus hire to Lake Victoria) and variable costs (costs that change with the number of participants, e.g., entry fees per person). – Identify sources of income and categories of expenditure relevant to a school club or society. – Recognise that a budget shortfall occurs when total expenditure exceeds total income, linking back to Amina's situation.
Skills	– Prepare a simple income-and-expenditure budget for a hypothetical Environmental Club trip to Lake Victoria, listing and totalling both fixed and variable costs. – Identify and articulate budget shortfalls by comparing planned income against planned expenditure. – Communicate financial reasoning clearly to peers using correct mathematical vocabulary (income, expenditure, surplus, deficit, fixed, variable).
Attitudes	– Appreciate the importance of careful financial planning before spending money, connecting to Amina's real-life challenge. – Demonstrate responsibility and integrity when handling shared club funds by recognising that every shilling must be accounted for. – Show curiosity and open-mindedness about questions that the budget raises — discounts, commissions, exchange rates — that will be explored in later lessons.
Key Inquiry Question	Why do we need a budget?
Purpose in Storyline	This opening lesson anchors the entire sub-strand by introducing Amina's KSh 4,500 dilemma. Learners immediately see that smart money decisions require planning, and they experience budget shortfalls first-hand through the Environmental Club trip scenario — creating the cognitive need to understand discounts, commissions, profit/loss, and exchange rates in subsequent lessons.
Safety Notes	No specific physical hazards. Remind learners to treat peer-shared financial information (e.g., amounts saved at home) with respect and sensitivity; do not pressure anyone to disclose personal financial details.

B. LESSON OVERVIEW

This is the opening lesson of the Commercial Arithmetic 1 sub-strand and serves as the anchor point for the entire unit storyline. The lesson begins by presenting Amina — a Form 4 student from Kisumu who has saved KSh 4,500 from selling mandazi at school — and her three-way money dilemma: contributing to the Environmental Club's end-of-term trip to Lake Victoria, buying a discounted smartphone at a Kondele electronics stall, and sending remaining money to her aunt in Uganda via mobile money. Before any instruction occurs, learners are asked to predict what Amina should do, surfacing their prior knowledge and intuitions about money management. This productive struggle creates the intellectual need for the mathematics that follows.

In the Observe phase, learners examine a realistic scenario card describing the Environmental Club's planned trip to Lake Victoria — including bus hire, park entry, lunch (ugali and fish), and a photography session at the lake shore — and sort the listed costs into fixed and variable columns. This concrete, Kenya-grounded activity makes the abstract distinction between fixed and variable costs immediately visible. Learners then total each column and compare the grand total against the club's available funds, discovering a shortfall that mirrors Amina's own predicament.

The Explain and Model Building phases guide learners to formalise the structure of an income-and-expenditure budget, articulate what a deficit means in practical terms, and pose their first questions on the Driving Question Board about what mathematical tools (discounts? commissions? exchange rates?) could help Amina — and the club — close the gap. By the end of the lesson, learners have collaboratively constructed a partial budget, identified a shortfall, and generated a rich set of unanswered questions that will propel the next five lessons forward.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Energy inputs for tilling a hectare of land Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Heat Budget of Planet Earth (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners read the Amina scenario (printed card or read aloud by teacher) and individually write their answer to the prompt: 'Amina has KSh 4,500. Write down, in any order, the three things she wants to do with it. Then predict: will KSh 4,500 be enough? Circle YES or NO and write one sentence explaining why you think so.' After writing, learners share predictions with a shoulder partner (Think-Pair). Three to four pairs share their reasoning with the class before any teaching takes place.	Distribute scenario cards (or read aloud twice, slowly). Say: 'Read carefully. Amina is a real student facing a real money problem. Before I teach you anything, I want to know what YOU think.' Allow 3 minutes of silent individual writing — do NOT prompt or hint. After writing, say: 'Turn to your shoulder partner. You have 90 seconds. Share your prediction and your reason.' After pair talk, cold-call EXACTLY 4 pairs using a random name stick: 'Kamau, what did your pair predict — and why?' Apply a minimum WAIT TIME of 12 seconds after each question before accepting answers. Record key prediction words on the board in a T-chart labelled 'YES — Enough' and 'NO — Not Enough'. Do not confirm or deny; say: 'Interesting. Let us keep these predictions visible. By the end of today you will know who was right — and why it matters.'	Think-Pair-Share Prediction: This strategy activates prior knowledge about money and spending before instruction, surfaces misconceptions (e.g., 'KSh 4,500 is a lot of money'), and creates cognitive dissonance that motivates the subsequent activity. Recording predictions publicly on the board makes student thinking visible and creates a reference point for revision at the lesson's end.	Teacher circulates during individual writing and looks for: (1) whether learners can correctly identify Amina's three goals (club trip, smartphone, mobile money transfer) — if fewer than half can do so, re-read the scenario aloud and ask learners to underline each goal; (2) the quality of reasoning in the YES/NO justification — are learners thinking mathematically (e.g., 'KSh 4,500 minus 20% is …') or only intuitively? Note which learners are already computing mentally — they may need extension prompts in later phases.
Observe Phase 📹 **VIDEO:** Energy inputs for tilling a hectare of land Source: Khan Academy (English	In groups of four, learners receive an 'Environmental Club Trip Scenario Card' listing eight cost items for a one-day trip from their school to Lake Victoria (Kisumu waterfront). The card lists: (A) Bus hire — KSh 3,200 regardless of group size; (B) National Museums	Distribute scenario cards and blank budget tables (two columns: Fixed Costs / Variable Costs, with rows for item name, unit cost, quantity, and total). Say: 'Your job right now is NOT to solve Amina's problem yet. Your job is to understand what a club budget	Categorisation-and-Calculation Table: Sorting costs into Fixed vs Variable before computing totals forces learners to make a conceptual decision (not just arithmetic) for each item. The structured table scaffolds mathematical communication and	Teacher looks for: (1) Correct classification — bus hire, agenda booklets, photography, first-aid kit as FIXED; entry, lunch, water, T-shirts as VARIABLE. If groups misclassify T-shirts (printed per person) as fixed, note this for the Explain phase discussion. (2)

- US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Heat Budget of Planet Earth (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	of Kenya entry — KSh 80 per learner; (C) Club chairperson's printed agenda booklets — KSh 150 flat; (D) Lunch: ugali, fried tilapia, and sukuma wiki — KSh 120 per learner; (E) Photography at the lakeshore — KSh 500 for the group; (F) Drinking water (500 ml bottles) — KSh 30 per learner; (G) First-aid kit replenishment — KSh 200 flat; (H) Club T-shirt printing — KSh 180 per learner. The club currently has 20 contributing members each paying KSh 300. Learners (i) sort all 8 costs into a two-column table: FIXED COSTS vs VARIABLE COSTS; (ii) calculate the total fixed cost; (iii) calculate the total variable cost for 20 learners; (iv) calculate grand total expenditure; (v) calculate total income; and (vi) determine whether there is a surplus or a deficit and by how much.	looks like by working through this Environmental Club trip. Work in your group of four. You have 10 minutes.' Set a visible timer. Circulate and ask probing questions — do NOT give answers: 'How did you decide that bus hire is fixed?' / 'What happens to the lunch cost if 5 more learners join?' / 'How is this connected to Amina's situation?' After 10 minutes, call groups to attention. Cold-call 3 different groups (use name sticks) to share: one group presents their Fixed Costs total, one presents Variable Costs total, one states whether there is a surplus or deficit. Apply 12-second WAIT TIME after each prompt. Write the agreed figures on the board. Expected answers: Fixed = KSh 3,200 + 150 + 500 + 200 = KSh 4,050; Variable = (80 + 120 + 30 + 180) × 20 = KSh 410 × 20 = KSh 8,200; Grand Total = KSh 12,250; Income = 20 × 300 = KSh 6,000; Deficit = KSh 6,250. Say: 'So the club is short by KSh 6,250. Sound familiar? Amina is also short. This is the problem we are going to solve — together — over the next lessons.'	ensures all group members engage with every item. Connecting the club's deficit back to Amina's shortfall anchors the mathematics in the phenomenon's storyline.	Correct multiplication of variable unit costs by 20 — watch for learners who add unit costs before multiplying (common error). (3) Correct identification of deficit (not surplus) — if a group concludes there is a surplus, this is a priority misconception to address in whole-class share-out. (4) Listen for groups spontaneously connecting the club's deficit to Amina's dilemma — these are high-leverage contributions to amplify.
Explain Phase 📹 **VIDEO:** [Energy inputs for tilling a hectare of land] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Heat Budget of Planet Earth (Flexbook)] Source: CK-12	Learners collaborate in the same groups to formalise their budget into a structured Income-and-Expenditure statement using a standard template drawn on the board by the teacher. They label columns correctly (Item | Amount KSh), write headings for INCOME and EXPENDITURE sections, and compute the NET POSITION (surplus or deficit). Each group then answers three 'So What?' questions in their exercise books: (1) 'What does a deficit mean for the club chairperson?' (2) 'If each	Draw the following on the board and explain each heading as you write it: 'INCOME: [source, amount]. TOTAL INCOME: ___. EXPENDITURE: Fixed Costs [list], Variable Costs [list]. TOTAL EXPENDITURE: ___. NET POSITION: Total Income − Total Expenditure = ___ (SURPLUS if positive, DEFICIT if negative).' Say: 'Copy this structure into your exercise book. Now re-write your Environmental Club budget into this format.' Give 4 minutes. Then pose the three 'So What?'	Structured Note-Making with Anchored Application: The formal Income-and-Expenditure template gives learners a reusable mathematical structure they will apply across the sub-strand. The three 'So What?' questions move learners from computation (what is the deficit?) to interpretation (what does it mean?) to connection (how does this explain Amina?), following the Explain-Apply-Connect sequence of sensemaking. Writing individual answers before group sharing	Teacher checks exercise books during individual writing for: (1) Correct NET POSITION formula and sign convention (deficit is negative or stated as 'short by'); (2) Q2 working — learners should compute 20 × 50 = KSh 1,000 additional income, then check: KSh 6,000 + 1,000 = KSh 7,000 < KSh 12,250, so NO, the deficit is NOT cleared (a common error is to forget the remaining KSh 5,250 gap); (3) Q3 — learners should explicitly name a number from Amina's scenario. If most learners

🔍 Search ARES for similar readings	of the 20 members paid KSh 50 more, would the deficit be cleared? Show your working.' (3) 'How does Amina's KSh 4,500 situation compare to the club's situation?' Learners write individual answers before group discussion.	questions verbally and write them on the board. Give 5 minutes for individual written answers. Cold-call 3 individual learners (not group reporters — pick quieter learners) for Q1, Q2, Q3 respectively. For Q2, say: 'Show us your working on the board' — hand the learner a chalk/marker. Apply 12-second WAIT TIME. For Q3, press deeper: 'Can you give me a number that shows how Amina's situation is similar?' (Expected: Amina also faces a deficit — she needs more money than she has for all three goals, just as the club's KSh 6,000 income falls KSh 6,250 short of the KSh 12,250 needed.) Close by saying: 'Today we have discovered that a budget is a mathematical plan. It reveals the truth about money — whether you have enough or not. Amina needs this truth before she can make any smart decision.'	ensures every learner constructs their own explanation.	cannot connect Q3 to a number, this indicates the phenomenon has not yet been internalised — repeat the Amina scenario card reading and ask: 'What three amounts does Amina need? Add them up.'
Driving Question Board (DQB) Creation 📹 **VIDEO:** Energy inputs for tilling a hectare of land Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Heat Budget of Planet Earth (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives one sticky note (or a small square of paper) and writes ONE question that Amina's story has raised in their mind that they do not yet know how to answer mathematically. Learners post their sticky notes on the class Driving Question Board — a large sheet of paper or a section of the board divided into four zones: DISCOUNTS & COMMISSIONS | PROFIT & LOSS | EXCHANGE RATES | OTHER QUESTIONS. Learners read each other's questions silently as they post them, and the class does a brief 'gallery walk' (60 seconds) around the board before returning to seats.	Say: 'We have started to understand why Amina — and the club — need a budget. But Amina's problem is bigger than a budget. She also faces a 20% discount she does not fully understand, a mobile-money commission she did not plan for, and an exchange rate that changed overnight. These are the mathematics we will build over the next lessons. Right now, I want YOUR questions — not my answers.' Distribute sticky notes. Say: 'Write ONE question that Amina's story has left unanswered in your mind. Make it a mathematics question if you can. You have 2 minutes.' After posting, point to the four zones: 'I have sorted the board into four areas. As you post your note, try to place	Driving Question Board (DQB) Launch: The DQB is the central sensemaking artefact of the entire unit. By requiring learners to generate questions rather than answers at this stage, the strategy positions learners as investigators rather than passive recipients. Sorting questions into thematic zones (discounts, commissions, exchange rates) previews the sub-strand structure, and the gallery walk builds collective ownership of the inquiry. Each subsequent lesson will return to the DQB to mark questions as 'answered' and add new ones.	Teacher reads all sticky notes during the gallery walk and notes: (1) The proportion of questions that are genuinely mathematical (e.g., 'How do you calculate 20% of something?') vs purely narrative (e.g., 'Why doesn't Amina just ask her parents?') — if fewer than half are mathematical, the class needs more scaffolding in Lessons 2–3 to connect context to computation; (2) Whether any learner has already articulated a question about exchange rates or commission — these learners can be assigned peer-teaching roles in later lessons; (3) Common misconceptions embedded in questions (e.g., 'Does 20% off mean you pay 20%?' — this surfaces a critical discount misconception to address in

		it in the zone that best fits your question. If it does not fit anywhere, put it in OTHER QUESTIONS.' Conduct the 60-second silent gallery walk. Then cold-call 4 learners to read one question from the board (not their own) and say why it matters to Amina's story. Apply 10-second WAIT TIME after each. Say: 'This board stays up for the whole unit. Every lesson, we will answer some of these questions — and probably add new ones. This is how mathematicians work: questions drive discovery.'		Lesson 2).
Model Building Phase 📹 **VIDEO:** [Energy inputs for tilling a hectare of land] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Heat Budget of Planet Earth (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners individually draw a simple 'Money Flow Diagram' for Amina in their exercise books. The diagram has one box labelled 'INCOME: KSh 4,500 (mandazi sales)' at the top, with three arrows flowing down to three boxes: (1) 'Environmental Club Trip Contribution — KSh ???', (2) 'Smartphone (20% discount from ???) — KSh ???', and (3) 'Mobile Money to Uganda — KSh ???'. Learners fill in any amounts they can estimate or guess right now and write a '?' in boxes they cannot yet fill. Below the diagram, learners write one sentence: 'Amina's biggest mathematical challenge in this plan is _______ because _______.' This is their first cumulative model — it will be revised and completed in each subsequent lesson as new concepts are learned.	Say: 'In this unit, you are going to build a model — a picture — of how Amina's money flows. Today you draw the skeleton. Each lesson, you will fill in more boxes with real mathematics. By the end of the unit, Amina's full financial picture will be complete — and you will have the tools to give her real advice.' Draw the skeleton diagram on the board as you explain: 'One box at the top: her income. Three arrows going down: three ways the money might leave. Label each arrow with what you know, and put a question mark where you are not sure yet.' Give 4 minutes for individual drawing. Then say: 'Write your one-sentence summary at the bottom: Amina's biggest challenge is _______ because _______.' Cold-call 3 learners to share their sentence. Apply 12-second WAIT TIME. Listen for learners who identify the budget shortfall as the core problem — affirm this: 'Exactly. The budget is the foundation. Without knowing if she has enough, every other decision is just guessing.' Close: 'Next lesson, we will give Amina —	Cumulative Skeleton Model: The Money Flow Diagram is the learner's personal model of the phenomenon that grows with each lesson. Starting with '???' in most boxes is intentional — it represents the current boundary of understanding and makes future learning feel purposeful (filling in the unknowns). This technique, adapted from NGSS modelling practices, makes the progression of understanding visible and concrete. The one-sentence summary develops the skill of articulating a mathematical problem in words — a key competency for the KICD assessment rubric.	Teacher collects or photographs 5–6 models (from a cross-section of ability levels) to review before Lesson 2. Look for: (1) Correct identification of three destinations for Amina's money; (2) Whether learners attempt any numbers — even rough estimates — or leave all boxes blank (blank boxes may indicate the scenario was not fully understood); (3) Quality of the one-sentence summary — strong responses name a specific mathematical concept (e.g., 'Amina's biggest challenge is the discount because she does not know how much KSh she saves'); weak responses remain vague (e.g., 'she does not have enough money'). Use this evidence to differentiate the opening activity of Lesson 2.

		and you — a precise mathematical tool for one of those question-mark boxes: the discount on the smartphone.' Point to the DQB: 'Some of your questions up there are about to get answered.'		

D. TEACHER REFLECTION
1. During the Predict Phase, what proportion of learners correctly identified all three of Amina's financial goals (club trip, smartphone discount, mobile money transfer)? If learners struggled, how might I simplify or re-present the scenario card for Lesson 2's opening review — perhaps by using a visual diagram rather than a prose paragraph?\n2. In the Observe Phase, which cost classifications caused the most disagreement in groups (e.g., T-shirt printing — fixed or variable?)? How can I use that productive disagreement as a teaching moment at the start of Lesson 2, and does it suggest I need to revisit the fixed/variable distinction before moving to discounts?\n3. How mathematically confident were learners' 'So What?' responses in the Explain Phase, particularly Q2 (testing whether an extra KSh 50 per member clears the deficit)? Did learners compute systematically or guess? What does this tell me about their readiness for percentage calculations in Lesson 2?\n4. Looking at the DQB sticky notes, which zone received the most questions — Discounts & Commissions, Profit & Loss, Exchange Rates, or Other? Does the distribution of questions align with the sequence of lessons I have planned, or do I need to reorder or emphasise certain concepts earlier?\n5. Were all learners able to draw a meaningful Money Flow Diagram, or did some produce only a box with Amina's name? What targeted support (a partially completed template, a peer model, or a structured sentence frame) should I prepare for those learners before Lesson 2 begins?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners examined the Environmental Club's trip budget for a one-day visit to Lake Victoria, sorting eight cost items (bus hire, entry fees, ugali-and-tilapia lunch, T-shirt printing, etc.) into fixed and variable categories, and calculated that the club's KSh 6,000 income falls KSh 6,250 short of the KSh 12,250 total expenditure needed.
What did I learn?	A budget is a mathematical plan comparing income (money coming in) and expenditure (money going out). Costs can be fixed (they do not change with the number of participants, e.g., bus hire) or variable (they increase with each additional participant, e.g., entry fees). When total expenditure exceeds total income, there is a deficit — a shortfall that requires a mathematical solution.
How does this explain the phenomenon?	Amina's confusion about her KSh 4,500 begins with the same problem the club faces: she has not yet written down all her expected costs and compared them to her income. Without a budget, she cannot know whether she truly has enough — or how much she is short — before she starts spending. This is why every Kenyan who wants to make a smart money decision, whether for a school trip, a smartphone purchase, or a cross-border transfer, must start with a budget.

LESSON 2 (40 minutes): Discounts and Percentage Discounts — How Much Does Amina Really Save at the Kondele Electronics Stall?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners use Amina's smartphone purchase at a Kondele electronics stall to calculate discount amounts and percentage discounts from marked prices, discovering why '20% off' does not always mean what it seems.
Knowledge	– Define discount as the reduction in the marked price of an item – State the formula: Discount = Marked Price − Sale Price – State the formula: Percentage Discount = (Discount ÷ Marked Price) × 100 – Distinguish between marked price and sale price in real shopping contexts
Skills	– Calculate the discount amount and sale price given a marked price and a percentage discount – Calculate the percentage discount given a marked price and a sale price – Apply discount calculations to judge whether a purchase is affordable within a budget
Attitudes	– Appreciate the importance of verifying advertised discounts with mathematics before spending – Show responsibility in making informed financial decisions as a consumer
Key Inquiry Question	Why are discounts and commissions necessary in trade?
Purpose in Storyline	Amina has spotted a second-hand smartphone at a Kondele stall advertised at 'KSh 3,800 — 20% off.' In Lesson 1 she built a budget and allocated KSh 1,500 to the Environmental Club trip. This lesson asks: after the 20% discount, what is the actual sale price of the phone, and can Amina afford it from her remaining savings? Learners role-play as shoppers and stall-keepers to build the mathematical tools they need to answer that question.
Safety Notes	No specific hazards. Ensure role-play remains respectful; remind learners that price negotiation scenarios should not mock or belittle any group.

B. LESSON OVERVIEW

This lesson is the second in an eight-lesson evidence-gathering sequence anchored on Amina's challenge of spending her KSh 4,500 wisely. Having established in Lesson 1 that a budget is a mathematical plan for income and expenditure, learners now zoom in on one specific line of Amina's budget: buying a second-hand smartphone at a Kondele electronics stall in Kisumu that is advertised at KSh 3,800 with a 20% discount. The central question is deceptively simple — how much does Amina actually pay? — but it requires learners to unpack what a percentage discount means, compute the discount amount, and determine the sale price. The lesson is structured around a role-play shopping scenario in which half the class acts as stall-keepers at "Kondele Electronics" and the other half acts as customers comparing prices.

The lesson moves from prediction (estimate the sale price before calculating) through structured observation of worked examples drawn from familiar Kenyan market contexts, to collaborative explanation using the discount and percentage discount formulas. By working through multiple scenarios — a maize grinder in a Nakuru hardware, a school bag in a Nairobi supermarket, and the Amina smartphone — learners develop fluency with the two-way relationship between discount amount and percentage discount. They learn both to find the sale price given the percentage, and to find the percentage discount given two prices.

Throughout, the anchoring phenomenon remains visible: every calculation is evaluated against Amina's remaining budget (KSh 4,500 − KSh 1,500 club contribution = KSh 3,000 available for the phone). The lesson ends with learners updating the class Driving Question Board and revising their cumulative financial model of Amina's KSh 4,500, replacing the estimated smartphone cost with the mathematically computed sale price.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Percent word problems: tax and discount Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** The Price of a Unit (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a small card showing: 'A second-hand smartphone is marked at KSh 3,800. The stall advertises 20% off. Estimate the sale price — do NOT calculate yet. Write your estimate on your card and hold it up.' Learners write a personal estimate, then share with a shoulder partner and agree on a joint prediction. The class hears three to four paired estimates read aloud before any computation begins.	Display the Amina scenario on the board: 'Amina sees a sign at Kondele Electronics: Smartphone — KSh 3,800. TODAY ONLY: 20% OFF!' Point to the sign and say, 'Before we touch a single number, I want your gut feeling. If something is 20% off KSh 3,800, roughly how much will Amina pay? Write it on your card — no calculators, no working, just your best estimate.' Allow WAIT TIME of 15 seconds for silent individual writing. Then say, 'Turn to your shoulder partner. You have 30 seconds — agree on one number between you.' After pairs confer, cold-call exactly 4 pairs using name sticks: 'Pair 1 — what was your estimate and why did you choose that number?' Record all four estimates on the board in a visible 'Our Predictions' column. Do NOT confirm or deny any estimate. Ask, 'Which estimate is closest to KSh 3,000 — the money Amina has left after her club contribution? Does that matter?' Allow 10 seconds of think time before taking one volunteer response.	Anchored Estimation — learners commit to a prediction before instruction, activating prior number sense about percentages and creating cognitive tension that motivates the Observe phase. Recording estimates publicly on the board creates accountability and a reference point to revisit at the end of the lesson.	Teacher scans held-up cards to gauge the range of estimates. Red flag: if most estimates are above KSh 3,500 (showing weak percentage intuition) or below KSh 2,500 (over-discounting), the teacher notes this for targeted clarification in the Explain phase. Green flag: estimates clustering around KSh 2,900–KSh 3,200 suggest reasonable proportional reasoning.
Observe Phase 📹 **VIDEO:** Percent word problems: tax and	Learners participate in a structured role-play called 'Kondele Market Day.' Six learners volunteer as stall-keepers; the rest are customers. Each stall-keeper	Before role-play begins, say: 'You are going to be shoppers and traders at Kondele Market. Stall-keepers, your job is to tell customers your marked price and	Role-Play with Structured Data Collection — the market role-play situates abstract percentage calculations in a lived Kenyan trading context, making the	Teacher looks for: (1) correct identification of which number is the marked price vs. sale price; (2) correct order of operations — multiply first, then subtract; (3)

discount Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** The Price of a Unit (PLIX) Source: CK-12 🔍 Search ARES for similar readings	receives a card showing a marked price and a discount percentage for a familiar Kenyan item (e.g., 'School bag — KSh 1,200, 15% off'; 'Packet of unga — KSh 180, 10% off'; 'Jiko — KSh 850, 25% off'). Customers visit stalls, record the marked price and discount on a simple two-column table in their exercise books, and — using only the information given — try to work out what they would actually pay. After 6 minutes of role-play, the whole class reconvenes and three stall-keepers share their item's details. The teacher works through the calculation for each item on the board step-by-step, narrating each operation.	your discount percentage — nothing else. Customers, your job is to work out what you would actually pay. Record everything in your table.' Distribute role cards. Circulate during the 6-minute role-play; listen for learners attempting multiplication versus subtraction errors. After reconvening, select stall-keeper 1 (school bag). Say, 'Tell us your item and its marked price.' Write on board: 'Marked Price = KSh 1,200. Discount = 15%.' Then narrate: 'Step 1 — I need to find 15% OF KSh 1,200. That means 15 divided by 100, multiplied by 1,200.' Write: Discount = (15/100) × 1,200 = KSh 180. 'Step 2 — I subtract the discount from the marked price.' Write: Sale Price = 1,200 − 180 = KSh 1,020. WAIT TIME 10 seconds. Cold-call one customer who visited that stall: 'Did you get the same answer? Where did your working differ?' Repeat for the jiko (KSh 850, 25% off). Then pose: 'Now let's apply this to Amina's smartphone — KSh 3,800 at 20% off. Work it out in your books. You have 90 seconds.' Cold-call 2 learners to share working on the board.	observation concrete and motivating. The two-column table (Marked Price | Sale Price) is a data-gathering scaffold that mirrors how a careful shopper — or Amina — would compare prices before deciding.	whether learners write the discount as a fraction of 100 or mistakenly subtract the percentage directly from the shilling amount (e.g., writing 3,800 − 20 = 3,780 — a common error to address). Teacher notes which learners complete the Amina calculation correctly within 90 seconds.
Explain Phase 📹 **VIDEO:** Percent word problems: tax and discount Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** The Price of a Unit (PLIX) Source: CK-12	Learners work in groups of three to complete a structured sensemaking task called 'The Two-Way Discount Table.' Each group receives a table with four rows. In rows 1–2, they are given the marked price and percentage discount and must find the sale price (forward calculation). In rows 3–4, they are given the marked price and the sale price and must find the percentage discount (reverse calculation). Row 4 is the Amina scenario restated in	Introduce the reverse problem by saying: 'So far we have gone in one direction — from percentage to sale price. But what if a trader tells you the price and you want to know the percentage discount? Let's find out.' Write on the board: 'Percentage Discount = (Discount ÷ Marked Price) × 100.' Annotate each term: 'Discount means the amount you save — that is Marked Price minus Sale Price.' Give groups 5 minutes for the table task. Circulate and ask probing	Two-Way Structured Inquiry Table — presenting both the forward (percentage → sale price) and reverse (sale price → percentage) calculations in a single table forces learners to understand the relationship between the quantities rather than memorising a single procedure. This mirrors NGSS Science and Engineering Practice 5 (Using Mathematics and Computational Thinking) by requiring learners to use mathematical representations to	Teacher checks group tables for: (1) correct formula application in both directions; (2) whether learners correctly compute the discount amount (Marked Price − Sale Price) before dividing — the most common error is dividing the sale price by the marked price rather than the discount; (3) whether explanations are written in words alongside calculations. Green flag: group correctly identifies Discount = KSh 760 and Percentage Discount = 20%. Red

🔍 Search ARES for similar readings	reverse: 'A smartphone's marked price is KSh 3,800. At the Kondele stall it was sold for KSh 3,040. What percentage discount was given?' Groups write full working in their exercise books, then one representative writes the group's answer for row 4 on the class board.	questions to any group stuck on the reverse calculation: 'What is the first thing you need to find — the discount or the percentage? So what do you calculate first?' After groups post their row-4 answers on the board, facilitate a whole-class comparison. If answers vary, say: 'We have two different answers here — KSh 760 saved, giving 20%, and one group saying 25%. Let's check both together.' Work through the correct solution publicly. WAIT TIME 15 seconds after posting the formula before asking: 'Can someone tell me in their own words what this formula is asking?' Cold-call 3 learners using name sticks. Explicitly connect to Amina: 'So when the stall advertises 20% off, Amina can now verify that claim herself with this formula — she does not have to trust the sign.'	solve problems in both directions.	flag: groups dividing 3,040 ÷ 3,800 and calling the result the discount percentage.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Percent word problems: tax and discount Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** The Price of a Unit (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each learner writes on a sticky note (or a torn scrap of paper) one new question this lesson has raised for them about Amina's situation. Examples might include: 'What if the stall negotiates — is the negotiated price a discount too?' or 'Does Amina pay the sale price or can the price still change with VAT?' Learners post their notes on the class DQB under the column heading 'Discounts.' They then read three notes posted by classmates and place a tick on any question they share. The three most-ticked questions are read aloud and circled as priority questions for future lessons.	Point to the DQB and say: 'In Lesson 1 we asked big questions about Amina's KSh 4,500. Today we have answered one piece of the puzzle — the smartphone discount. But I am sure new questions have come up. Take 60 seconds to write one genuine question you still have — something this lesson has made you more curious about.' WAIT TIME 60 seconds. As learners post notes, say: 'Walk around, read what your classmates have written, and tick any question you share.' After 2 minutes, identify the top three ticked questions and read them aloud: 'Our class most wants to know: [read question 1], [read question 2], [read question 3]. We will come back to these as we continue Amina's story.' Connect explicitly to the driving	Question Sorting and Prioritisation — the DQB functions as a public record of the class's growing understanding. Allowing learners to tick shared questions builds collective ownership of the inquiry and surfaces genuine confusion for the teacher to address in future lessons. This mirrors NGSS Practice 1 (Asking Questions) by positioning learner-generated questions as drivers of the investigation.	Teacher reads all posted notes before the next lesson. Look for: (1) questions that reveal misconceptions (e.g., 'Is the 20% taken off the sale price again if there is another discount?' — indicating confusion about cumulative discounts); (2) questions that anticipate future lessons (e.g., questions about commission or exchange rates showing learners are connecting ahead); (3) whether all learners posted a note — those who did not are followed up individually.

		question: 'All of these questions are really asking the same big thing — how can Amina use mathematics to make smart money decisions? Today we added one tool: the discount formula.'		
Model Building Phase 📹 **VIDEO:** Percent word problems: tax and discount Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** The Price of a Unit (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the financial model of Amina's KSh 4,500 that they began in Lesson 1 (a simple allocation diagram or table in their exercise books). They now update the smartphone line: replace the rough estimated smartphone cost with the mathematically computed sale price of KSh 3,040 (20% off KSh 3,800). They recalculate Amina's remaining balance after the smartphone purchase: KSh 3,000 − KSh 3,040 = −KSh 40. Learners record this new finding — Amina is KSh 40 short — and write one sentence explaining what Amina should do next, referencing the mathematics from today.	Say: 'Open your financial model from Lesson 1. Let's update it with what we learned today.' Draw the model structure on the board: 'Total savings: KSh 4,500. Minus club contribution: KSh 1,500. Remaining: KSh 3,000. Smartphone sale price: KSh 3,040.' Pause and ask: 'What does this tell us about Amina's plan?' WAIT TIME 15 seconds. Cold-call 2 learners. Expected answer: Amina cannot afford the phone at this price — she is KSh 40 short. Say: 'Write that finding in your model. Then write one sentence: What should Amina do? Use the word discount in your answer.' Give 2 minutes for individual writing. Cold-call 3 learners to read their sentence. Highlight any response that suggests Amina negotiate a higher discount, save more, or reconsider the club contribution — all valid mathematical responses. Close by saying: 'Our model is getting more detailed. Next lesson we will look at commissions — the agent's charge Amina will face when she sends money to Uganda. That will change the model again.'	Cumulative Model Revision — learners maintain a running financial model across all eight lessons. Each revision makes the model more accurate and more complex, mirroring how scientists update models as new evidence is gathered (NGSS Practice 2: Developing and Using Models). The moment of discovering Amina is KSh 40 short creates productive disequilibrium that motivates continued inquiry.	Teacher circulates to check updated models. Look for: (1) correct subtraction yielding a deficit of KSh 40; (2) whether learners' written sentence shows conceptual understanding (e.g., 'Amina should ask the trader for a higher discount or choose a cheaper phone') rather than procedural repetition; (3) whether the model layout is becoming more organised and labelled. Collect 5 exercise books at random to review model quality before Lesson 3.

D. TEACHER REFLECTION
1. During the Predict phase, what was the range of estimates? Did learners with stronger proportional reasoning in Lesson 1 (budget fractions) also estimate the discount more accurately — and if so, how can you leverage that connection explicitly in Lesson 3?\n2. In the role-play, which item contexts (unga, school bag, jiko) generated the most engagement and the most mathematical errors? What does this tell you about which real-world contexts are most meaningful to your learners?\n3. When learners encountered the reverse calculation (finding percentage discount from two prices), what was the most common error? Did you address it publicly, and did

your intervention change learners' understanding within the lesson?\n4. How many learners updated their financial model correctly to show Amina's KSh 40 deficit? Were there learners who still had incorrect totals from Lesson 1's budget — and how will you address cumulative errors before those compound in later lessons?\n5. Looking at the DQB sticky notes, which questions surprised you most, and how will you sequence the remaining six lessons to ensure those questions are answered by the end of the unit?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed, through a role-play shopping scenario at a simulated Kondele electronics stall, that a '20% off' sign reduces the marked price of KSh 3,800 to a sale price of KSh 3,040 — a saving of KSh 760 — and that this saving is computed by multiplying the marked price by the percentage expressed as a fraction of 100, then subtracting.
What did I learn?	Discount = Marked Price − Sale Price; Percentage Discount = (Discount ÷ Marked Price) × 100. These two formulas work in both directions: given a percentage discount you can find the sale price, and given both prices you can verify the percentage discount a trader claims to be offering.
How does this explain the phenomenon?	Amina's '20% off' smartphone deal reduces the price from KSh 3,800 to KSh 3,040, meaning she needs KSh 3,040 from her remaining KSh 3,000 — leaving her KSh 40 short. Mathematics reveals that the advertised discount is real but still leaves Amina unable to complete the purchase without adjusting another part of her budget, showing why verifying discounts with arithmetic — rather than trusting a sign — is essential for smart money decisions.

LESSON 3 (40 minutes): Discounts and Percentage Discounts (Part 2 / Practice) — Which Deal Gives Amina Better Value?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners practise and consolidate percentage discount calculations across varied Kenyan retail contexts, compare competing discount offers, and determine which deal gives better value — bringing Amina closer to a smart purchasing decision at the Kondele electronics stall.
Knowledge	– Discount = Marked Price − Sale Price; Percentage Discount = (Discount ÷ Marked Price) × 100 – A higher percentage discount does not always mean a greater absolute saving — the marked price matters – Sale price can be calculated directly as Sale Price = Marked Price × (1 − Percentage Discount ÷ 100) – Comparing deals requires converting all offers to the same unit (either absolute saving or percentage discount)
Skills	– Calculate discount and percentage discount fluently in varied Kenyan retail scenarios (Naivas Weekend Specials, mitumba clothing, school-uniform shops) – Compare two or more discount offers and justify which gives better value using mathematical reasoning – Present and defend a discount comparison to peers using clear mathematical language
Attitudes	– Develop critical consumer awareness — questioning advertised 'deals' rather than accepting them at face value – Appreciate mathematics as a practical tool for protecting oneself from misleading pricing in everyday Kenyan markets
Key Inquiry Question	Why are discounts and commissions necessary in trade?
Purpose in Storyline	In Lesson 2 Amina learned the basic discount formulas by focusing on one item — the smartphone at the Kondele stall. In Lesson 3 she (and we) face a harder real-world challenge: multiple competing deals. Naivas is advertising a 'Weekend Special — 25% off all electronics.' A mitumba stall near Kondele is selling a similar second-hand phone at KSh 3,200 with 'KSh 400 off.' Which is actually cheaper? Which percentage discount is bigger? By working through varied contexts and presenting their comparisons to peers, learners consolidate the formulas from Lesson 2 and build the consumer-mathematics reasoning that will underpin Lessons 4–8 on commissions, profit/loss, and currency exchange.
Safety Notes	No specific hazards. Ensure peer-discussion groups are inclusive and that no learner is singled out negatively during price-comparison presentations.

B. LESSON OVERVIEW

This lesson is the consolidation and application lesson for percentage discounts — the 'practice with variation' step in the evidence-gathering sequence. Having established the core formulas in Lesson 2, learners now encounter three distinct Kenyan retail contexts that deliberately vary the structure of the discount problem: (1) a Naivas Supermarket Weekend Special that gives the percentage discount and asks for the sale price; (2) a mitumba clothing stall in Gikomba that gives the absolute discount and asks for the percentage; and (3) a school-uniform shop in Kisumu that presents two competing offers and asks learners to determine which is better value. These three scenario types mirror the full range of discount problems that appear in KICD-aligned assessments and, more importantly, in real Kenyan market life.

The lesson is structured around a Predict–Observe–Explain–DQB–Model cycle. In the Predict phase learners commit — before any calculation — to gut-feel rankings of which deal is best, surfacing and then productively disrupting the common misconception that 'a bigger number in the advertisement always means a better deal.' In the Observe phase learners work in pairs through a structured worksheet of three Kenyan scenarios, recording both absolute discount and percentage discount so they can compare apples with apples. In the Explain phase, selected pairs present their comparisons to the class and the teacher orchestrates a discussion that surfaces the key conceptual move: to compare fairly, you must convert all offers to the same unit.

By the end of the lesson every learner can calculate discount and percentage discount in at least two different scenario structures, compare competing offers, and articulate in plain language why one deal gives better value — skills Amina needs before she can decide whether to buy the Kondele phone or wait for the Naivas sale, and skills that will be built upon in Lessons 4 and 5 when commissions and profit/loss enter the picture.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Percent word problems: tax and discount Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** The Price of a Unit (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown three deal cards displayed on the board (or read aloud by the teacher): Deal A — Naivas Weekend Special: a KSh 8,000 blender, 25% off; Deal B — Gikomba mitumba stall: a jacket originally KSh 1,200, now KSh 400 off; Deal C — Kisumu school-uniform shop: two schoolbags — Bag X at KSh 1,500 with 10% off, Bag Y at KSh 2,000 with KSh 250 off. Without doing any calculation, each learner individually writes in their exercise book: 'I think Deal ___ saves the most money in shillings. I think Deal ___ has the biggest percentage discount. I think Amina should buy ___.' Learners then share their predictions with a shoulder partner for 60 seconds.	Display the three deal cards clearly — write them on the board or read each one twice slowly. Say: 'Before we do a single calculation, I want your gut feeling. Look at these three deals. In your books, write your three predictions — biggest shilling saving, biggest percentage discount, and what Amina should do. You have 90 seconds. Go.' [WAIT — allow full 90 seconds of silent individual thinking and writing.] After 90 seconds: 'Tell your shoulder partner your predictions. Explain your reasoning in one sentence each.' [Allow 60 seconds of paired talk.] Cold-call 3 learners — one from each side of the room and one from the back: 'Akinyi, which deal gives the biggest shilling saving — and why do you think that without calculating?' After each response, ask: 'Does anyone predict differently? Raise your hand.' Do NOT confirm or correct predictions yet. Say: 'Hold those predictions. We are about to find out who is right — and the answer might surprise you.'	Prediction-with-Justification: Learners commit to a position and articulate their reasoning before instruction. This activates prior knowledge from Lesson 2, surfaces the misconception that 'bigger number = better deal,' and creates cognitive dissonance that motivates engagement with the Observe phase. Recording predictions in writing ensures accountability and provides a visible record for learners to revise.	Teacher scans the room during the 90-second write time — look for learners who are not writing (may need the scenario re-read) and for patterns in raised hands when asking 'who predicts differently.' Listen specifically for whether learners are reasoning about absolute amounts versus percentages — this reveals the depth of Lesson 2 transfer. If the majority of the class predicts the same deal for both 'biggest saving' and 'biggest percentage,' note this as a misconception to address explicitly in the Explain phase.
Observe Phase 📹 **VIDEO:** Percent word	Learners work in pairs on a structured three-scenario worksheet. Each scenario is set in a Kenyan retail context and	Distribute the worksheet and say: 'Work with your partner. For each scenario, show every step of your working — do not jump to the	Structured Pair Investigation with Graduated Scenarios: The three scenarios are deliberately sequenced — Scenario 1 gives	While circulating, look for: (1) Correct formula set-up but arithmetic errors — indicate the step is wrong without correcting it

problems: tax and discount Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** The Price of a Unit (PLIX) Source: CK-12 🔍 Search ARES for similar readings	requires them to: (a) calculate the absolute discount (KSh saved), (b) calculate the percentage discount, and (c) state the final sale price. Scenario 1 — Naivas Weekend Special: A Tecno Spark smartphone has a marked price of KSh 9,500 and is advertised at '20% off.' What is the discount in shillings? What is the sale price? Scenario 2 — Gikomba mitumba stall: A second-hand pair of Reebok trainers has a marked price of KSh 1,800. The seller offers KSh 360 off. What is the percentage discount? Scenario 3 — Comparison (Kisumu uniform shops): Shop Pendo charges KSh 2,400 for a school blazer and offers 15% off. Shop Fahari charges KSh 2,600 for the same blazer and offers KSh 450 off. Which shop gives a lower sale price? Which offers a bigger percentage discount? After completing the worksheet, pairs record their answers in a summary table with columns: Marked Price | Discount (KSh) | Percentage Discount | Sale Price.	answer. Use the formulas we established in Lesson 2: Discount = Marked Price minus Sale Price, and Percentage Discount = (Discount divided by Marked Price) times 100.' [WAIT — circulate immediately. Do not stand at the front.] Circulate for 10–12 minutes. At each pair, look at their working, not just their answer. Use these targeted prompts — do not give answers: For pairs who are stuck on Scenario 1: 'What does 20% of KSh 9,500 mean? Can you write 20% as a decimal?' For pairs who conflate absolute and percentage discount in Scenario 3: 'You have told me Shop Pendo saves more shillings. Is that the same as saying it has a bigger percentage discount? How would you check?' After 12 minutes, cold-call 3 pairs to share one answer each — Pair A: Scenario 1 discount in KSh; Pair B: Scenario 2 percentage discount; Pair C: which shop in Scenario 3 gives a lower sale price. Write each answer on the board as it is shared. Say: 'We will check these in our next phase.'	percentage, find price (direct application); Scenario 2 gives absolute discount, find percentage (reverse application); Scenario 3 requires comparison across two offers (synthesis). This gradation ensures all learners succeed at Scenario 1 while being challenged by Scenario 3. Recording in a summary table makes the comparison visible and prepares the Explain-phase discussion.	('Check your multiplication on step 2'). (2) Learners who compute percentage discount for Scenario 3 as (450 ÷ 2,400) — they have used the wrong marked price; redirect them to 'Which blazer does KSh 450 come off?' (3) Pairs who correctly find that Shop Pendo has a lower sale price (KSh 2,040) versus Shop Fahari (KSh 2,150) AND a bigger percentage discount (15% vs 17.3%) — note these pairs for Explain-phase cold-calling. Target: at least 70% of pairs complete Scenarios 1 and 2 correctly within 12 minutes.
Explain Phase 📹 **VIDEO:** Percent word problems: tax and discount Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** The Price of a Unit (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The teacher facilitates a whole-class discussion anchored on the three scenarios. Three selected pairs each present their working for one scenario on the board (or by reading aloud with the teacher scribing). After each presentation the class has 60 seconds to silently check their own working and mark it correct or incorrect. In the final 8 minutes, learners individually tackle a 'Connecting to Amina' problem: 'Amina saw the Kondele stall smartphone at KSh 9,500 marked price with 20% off (from Lesson 2). She now hears that Naivas has the same phone at	Invite Pair A to present Scenario 1. After they finish, ask the class: 'Thumbs up if your answer matches. Thumbs sideways if you got a different answer.' Address discrepancies before moving on. Then say: 'Here is the key question for Scenario 1 — did the 20% discount save more or fewer shillings than you predicted? Turn and tell your partner.' [WAIT 15 seconds.] Cold-call 2 learners. For Scenario 3 (the comparison), after Pair C presents, ask: 'Scenario 3 is the one that should make us think hardest. Shop Fahari charges a higher marked price AND gives a	Orchestrated Class Discussion with Cognitive Conflict Resolution: By displaying predictions from the Predict phase alongside calculated results, the teacher deliberately creates and then resolves cognitive conflict — particularly around the Scenario 3 result where a higher percentage discount coexists with a higher sale price. This is the core conceptual move of the lesson. The 'Connecting to Amina' problem then requires learners to apply this insight independently, anchoring the sensemaking back to the driving phenomenon.	During the 'Connecting to Amina' problem, circulate and look for: (1) Correct calculation — Kondele sale price = KSh 9,500 × 0.80 = KSh 7,600; Naivas sale price = KSh 10,200 × 0.75 = KSh 7,650. Kondele is cheaper by KSh 50, despite a smaller percentage discount. (2) Learners who choose Naivas because '25% is bigger' without calculating the sale price — this is the target misconception; mark these books with a follow-up prompt. (3) Quality of the one-sentence recommendation — look for mathematical justification ('Kondele is better because the

	KSh 10,200 with 25% off. Which deal gives a lower sale price? Which has a bigger percentage discount? Which should Amina choose — and why?' Learners write a full worked solution and a one-sentence recommendation for Amina in their exercise books.	bigger percentage discount — yet Shop Pendo gives a lower final price. How is that possible?' [WAIT 20 seconds — this is a rich conceptual question.] Cold-call 3 learners. Scribe their reasoning on the board. Then say: 'This is the mathematical lesson for all of us as consumers: a bigger percentage does not always mean you pay less. You must always calculate the final sale price.' Now display the 'Connecting to Amina' problem. Say: 'Amina needs YOUR help. Work this out individually in silence — full working, every step. You have 6 minutes.' [WAIT — circulate and observe.] After 6 minutes, cold-call one learner to read their recommendation for Amina. Ask: 'Does anyone disagree with this recommendation? Raise your hand and explain.'		sale price of KSh 7,600 is KSh 50 less than Naivas') rather than vague statements ('Kondele is cheaper'). At least 80% of learners should arrive at the correct sale prices for both options.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Percent word problems: tax and discount] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [The Price of a Unit (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Each learner writes one sticky note (or a slip of paper) responding to the prompt: 'What does today's lesson help you answer about Amina's situation — and what new question does it raise?' Learners bring their notes to the class Driving Question Board. The teacher reads several aloud and clusters them under two headings: 'We Can Now Explain…' and 'We Still Wonder…'. Learners then update their personal DQB tracking sheet by ticking off 'Calculate and compare percentage discounts' and adding any new personal questions raised by today's lesson.	Say: 'Take one slip of paper. Write two things: first, one thing today's lesson now lets you explain about Amina's money decisions — connect it directly to her KSh 4,500 and the phone at Kondele. Second, write one new question today's lesson has raised for you — something you still wonder about.' [Allow 3 minutes of silent individual writing. WAIT — do not rush this.] Invite learners to bring notes to the board. Read 4–5 notes aloud, paraphrasing: 'Kavata says she can now explain why 20% off a cheaper phone can cost less than 25% off a more expensive one — that goes under We Can Now Explain.' 'Omondi wonders: if the mobile-money agent takes a commission, is that like a reverse discount — a deduction Amina doesn't want?	Driving Question Board as a Living Artefact: The DQB is not a one-time activity but a cumulative record of the class's growing understanding. By physically placing notes and clustering them, learners see that science and mathematics knowledge builds iteratively — each lesson resolves some questions and opens new ones. Linking Omondi's commission question to Lesson 4 creates a narrative thread that maintains curiosity and storyline coherence.	Collect and read all DQB slips after class. Look for: (1) 'We Can Now Explain' notes that correctly articulate the key insight (must calculate sale price, not just compare percentages). (2) 'We Still Wonder' notes that reveal productive next-step questions — commission, exchange rates, profit/loss — these inform whether the storyline sequencing is working. (3) Slips that reveal persistent misconceptions ('a bigger percentage always saves more') — these learners need a brief 1-on-1 or small-group follow-up at the start of Lesson 4.

		Great question — that goes under We Still Wonder.' Point to the board: 'We are building this board lesson by lesson. Notice how our understanding of Amina's problem is growing.'		
Model Building Phase 📹 **VIDEO:** [Percent word problems: tax and discount] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [The Price of a Unit (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Learners open their cumulative 'Amina's Money Model' — a visual diagram started in Lesson 1 that maps how Amina's KSh 4,500 flows. In Lesson 2 they added the discount arrow on the smartphone purchase. Today they revise and extend that arrow: they calculate the actual sale price of the Kondele phone (KSh 7,600) and write it on the model, note that Amina's KSh 4,500 is still not enough (she is KSh 3,100 short), and add a new annotation: 'Comparing deals — the lower percentage discount can still give the lower price.' Learners also annotate which shop they would recommend to Amina and write a one-sentence mathematical justification next to the model arrow.	Say: 'Take out your Amina's Money Model from Lesson 1. Find the arrow you drew in Lesson 2 showing the discount on the Kondele smartphone. Today we are going to make that model more precise and more useful.' Display the model template on the board and demonstrate: 'In Lesson 2 we knew the phone was KSh 9,500 with 20% off. We now know the sale price is exactly KSh 7,600. Write that on your arrow. Amina has KSh 4,500 — annotate how far short she is.' [WAIT 2 minutes for learners to update.] 'Now add today's new insight as a label on or near the discount arrow — something that would help Amina as a consumer. What would you write in ten words or fewer?' [WAIT 15 seconds.] Cold-call 3 learners to share their label. Write the best one on the board model: for example, 'Always calculate final price — do not trust the percentage alone.' Say: 'In our next lesson, Amina discovers the mobile-money agent takes a commission. How will that change this model? Hold that thought.'	Cumulative Model Revision: The model is not redrawn from scratch but revised — this mirrors how scientific and mathematical models work. Each revision makes the model more accurate and more nuanced. Annotating the Kondele phone sale price with precision (KSh 7,600, not 'about KSh 7,000') develops quantitative precision as a habit of mind. The teacher's closing prompt ('how will commission change this model?') creates a forward-looking hook for Lesson 4.	Walk around and look at learner models during the 2-minute update. Check: (1) Is the sale price written correctly as KSh 7,600? (2) Is the shortfall annotated (KSh 9,500 × 0.80 = KSh 7,600; KSh 7,600 − KSh 4,500 = KSh 3,100 short)? (3) Is the consumer-insight label mathematically accurate and connected to the lesson's key idea? Collect 5–6 models to review after class as a quick written assessment sample. Return them with brief written feedback before Lesson 4.

D. TEACHER REFLECTION

1. During the Predict phase, did learners' gut-feel predictions reveal the target misconception (bigger percentage = better deal)? If the majority predicted correctly without calculating, how will you deepen the challenge for this class in future discount lessons?

2. In the Observe phase, which of the three scenario types caused the most difficulty — the direct calculation (Scenario 1), the reverse calculation (Scenario 2), or the comparison (Scenario 3)? What does this tell you about which skill needs more practice before the topic assessment?

3. When you orchestrated the Explain-phase discussion around Scenario 3 (Shop Pendo vs Shop Fahari), did learners articulate the conceptual insight ('a higher

percentage off a higher price can still leave you paying more') in their own words, or were they echoing your phrasing? How might you elicit more authentic mathematical reasoning next time? 4. Review the 'Connecting to Amina' individual written responses. What proportion of learners correctly calculated both sale prices AND gave a mathematically justified recommendation? What intervention will you plan for those who chose Naivas because '25% is bigger'? 5. Looking at the DQB slips collected, did learners' 'We Still Wonder' questions anticipate the next lesson topics (commission, profit/loss, exchange rates)? If not, what aspects of the Amina storyline need to be made more salient so learners feel the narrative pull toward the next evidence-gathering lesson?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners worked through three Kenyan retail discount scenarios — a Naivas Weekend Special, a Gikomba mitumba stall, and a Kisumu school-uniform comparison — and discovered that a higher advertised percentage discount does not always result in a lower sale price. In the Amina comparison problem, the Kondele phone (20% off KSh 9,500 = KSh 7,600) was KSh 50 cheaper than the Naivas phone (25% off KSh 10,200 = KSh 7,650), despite having a smaller percentage discount.
What did I learn?	Discount = Marked Price − Sale Price; Percentage Discount = (Discount ÷ Marked Price) × 100; Sale Price = Marked Price × (1 − Percentage Discount ÷ 100). To compare two discount deals fairly, you must calculate the actual sale price for each — you cannot simply compare percentage figures. A deal with a smaller percentage discount on a lower marked price may cost you less than a deal with a larger percentage discount on a higher marked price.
How does this explain the phenomenon?	Amina's instinct to chase the biggest percentage discount (25% at Naivas) would actually cost her KSh 50 more than buying at Kondele (20% off). This shows that the '20% off' advertisement on the Kondele stall is not misleading — it is genuinely the better deal for her, even though the percentage is smaller. For Amina to make a smart decision about her KSh 4,500, she must calculate the final sale price every time, not just compare the percentage figures on the advertisement boards.

LESSON 4 (40 minutes): Commission and Percentage Commission — Who Gets Paid When Amina Sends Money to Uganda?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners investigate how commissions work in real Kenyan financial transactions — from M-PESA agents in Kericho to market brokers in Kondele — so that Amina can understand how much of her KSh 4,500 will be deducted when she sends money to her aunt in Uganda.
Knowledge	– Define commission as an amount paid to an agent or intermediary for facilitating a sale or financial transaction. – Explain percentage commission as the commission expressed as a fraction of the transaction value, multiplied by 100. – Identify real-life Kenyan contexts where commissions are charged: M-PESA/Airtel Money agents, insurance brokers, real estate agents, market brokers, and mobile-money transfer services.
Skills	– Calculate the commission amount given a transaction value and a commission rate. – Calculate the percentage commission given the commission amount and the transaction value. – Determine the net amount received or paid after a commission is deducted from a transaction.
Attitudes	– Appreciate that understanding commission protects consumers from being overcharged and empowers them to make fair financial decisions. – Value transparency and honesty in financial transactions as a form of responsible citizenship.
Key Inquiry Question	Why are discounts and commissions necessary in trade?
Purpose in Storyline	Amina has already wrestled with discounts on the smartphone; now she must send the remaining money to her aunt in Uganda via a mobile-money agent. This lesson reveals that the agent charges a commission — meaning Amina's aunt will receive less than Amina sends unless Amina factors the commission into her plan. Calculating commissions and percentage commissions gives Amina (and every Kenyan) a mathematical tool to understand what intermediaries earn and how much the transaction truly costs.
Safety Notes	No specific hazards. Role-play money transactions use paper dummy currency only; remind learners not to share real personal financial information during simulations.

B. LESSON OVERVIEW

This lesson is the fourth in the evidence-gathering sequence and directly advances the anchoring phenomenon: Amina from Kisumu wants to send part of her remaining KSh 4,500 to her aunt in Uganda through a mobile-money agent, but the agent charges a commission. Learners first encounter the concept of commission through a relatable Kenyan scenario — a Kericho M-PESA agent who earns KSh 30 for every KSh 1,000 sent — and brainstorm other real contexts where commissions are paid (insurance agents, real estate brokers, Kondele market produce brokers, mobile-money kiosks). They then calculate commission amounts and percentage commissions from given transaction values using the formulas: Commission = Commission Rate × Transaction Value ÷ 100, and Percentage Commission = (Commission ÷ Transaction Value) × 100.

The lesson's central activity is a structured role-play in which pairs of learners act as "mobile-money agents" and "customers," using laminated dummy-currency cards. Each pair processes three transactions of different values and calculates the commission and percentage commission for each. This experiential, embodied learning strategy ensures that the abstract formula is grounded in the lived reality of a transaction. Learners also work backwards — given the commission amount and the transaction value, they determine what percentage the agent is charging — building both procedural fluency and conceptual understanding.

In the Explain and Model Building phases, learners connect commission to their earlier learning about discounts: both discount and commission are deductions expressed as percentages, but they fall on different parties (the buyer benefits from a discount; the intermediary earns a commission). By the end of the lesson, learners can clearly state what Amina's mobile-money agent earns and how much her aunt in Uganda will actually receive — a direct, satisfying answer to the phenomenon. The Driving Question Board is updated with new evidence about the hidden costs in financial transactions.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Genetics vocabulary Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Repaying Money (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners read a short scenario card on their desks: 'Amina walks to a mobile-money kiosk near Kondele market. She wants to send KSh 800 to her aunt in Uganda. The agent says: I will charge you a small fee for this service. Amina pays KSh 824. How much did the agent keep, and why?' Individually, learners write a prediction in their exercise books: (a) how much money did the agent keep? (b) what do you think that amount is called? (c) is this fair — should agents charge for their service? They are not expected to know the term yet; predictions are based on intuition and prior experience with transactions.	Distribute scenario cards face-down. Say: 'Before we flip the card, I want you to think about the last time someone in your family sent money using M-PESA or Airtel Money. Did the recipient get exactly the amount that was sent?' [WAIT 10 seconds for quiet reflection.] Say: 'Now flip the card and read silently.' [Allow 60 seconds.] Say: 'Write three predictions in your book — do not talk to your neighbour yet. You have 2 minutes.' After 2 minutes, cold-call 3 learners: 'Akinyi, tell me your prediction for (a).' 'Otieno, do you agree or disagree with Akinyi — why?' 'Wanjiku, what word did you use for the amount the agent kept?' Record all three vocabulary suggestions on the board under the heading: 'Our words for it today.' Do NOT correct or validate yet — affirm thinking: 'Interesting — we will test all these ideas.'	Predict-Before-Instruction (Elicitation of Prior Knowledge): by asking learners to commit a written prediction before any instruction, the teacher surfaces existing mental models about financial transactions. This strategy ensures that new vocabulary (commission) is introduced into a space already primed with real-world meaning, making the formal definition stick rather than float abstractly.	Teacher scans written predictions as learners write — look for: (1) correct arithmetic (KSh 824 − KSh 800 = KSh 24 kept by agent) as evidence of number sense; (2) use of words like 'fee,' 'charge,' 'cut,' or 'commission' as evidence of prior financial literacy; (3) reasoning about fairness — qualitative responses reveal attitude formation. Flag learners who write 'the agent kept nothing' for targeted questioning during the Observe phase.
Observe Phase 📹 **VIDEO:** Genetics vocabulary Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners participate in a structured role-play called 'Kericho Kiosk.' The class is divided into pairs; one learner is the 'Agent' and the other is the 'Customer.' Each pair receives a Role-Play Transaction Card with three transactions printed on it. Transaction 1: Customer sends KSh 500; agent charges KSh 15. Transaction 2:	Distribute Role-Play Transaction Cards and dummy-currency sets. Say: 'You are now a real mobile-money agent at a kiosk in Kericho — like the thousands of M-PESA agents across Kenya. Your job is to process the transaction AND keep your commission. Customer: count out the money you are handing over. Agent: count out	Embodied Role-Play with Data Table Construction: physical enactment of the transaction (counting dummy notes, handing money across a desk) makes the abstract concept of commission tangible and memorable. Building the data table during the role-play models the NGSS Science and Engineering Practice of 'Analysing	Walk the room during role-play and check three things: (1) Are both partners recording the correct amounts in the table — spot-check at least 5 pairs? (2) Do learners correctly subtract the commission to find what the recipient gets (e.g., KSh 500 − KSh 15 = KSh 485)? (3) In written observations, do learners identify a proportional

📖 **HTML:** Repaying Money (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Customer sends KSh 2,000; agent charges KSh 60. Transaction 3: Customer sends KSh 4,500 (Amina's full savings); agent charges KSh 135. Learners act out each transaction using dummy-currency cards (paper cut-outs of KSh notes), physically counting out the money the agent retains. After each transaction, both partners record in a table: Transaction Value | Commission Amount | Amount Recipient Gets. At the end of all three transactions, learners look at their table and write one observation: 'What do you notice about how the commission amount changes as the transaction value increases?'	exactly what you are keeping.' Circulate and narrate: 'I can see Chebet is counting carefully — good. Hassan, how much did your agent keep in Transaction 1?' [WAIT 10 seconds.] After all pairs finish, pause the role-play. Say: 'Look at your table. What is the relationship between the transaction value and the commission amount? Write one sentence — do not discuss yet.' [Allow 90 seconds.] Cold-call 4 learners to share observations. Write the three commission amounts (KSh 15, KSh 60, KSh 135) and corresponding transaction values (KSh 500, KSh 2,000, KSh 4,500) in a two-column table on the board. Ask: 'Can anyone see a pattern — what fraction or percentage is the commission each time?' [WAIT 15 seconds.] Cold-call 2 more learners. Guide them to notice: 15/500 = 60/2000 = 135/4500 = 0.03 = 3%.	and Interpreting Data' — learners do not just receive a formula; they construct the pattern from evidence.	relationship ('the more you send, the more commission') or a constant-ratio relationship ('commission is always 3%')? The latter shows deeper proportional reasoning and should be highlighted during the Explain phase.
Explain Phase 📹 **VIDEO:** Genetics vocabulary Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Repaying Money (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners transition from observation to formalisation. The teacher introduces the two core formulas and learners write them in their books with colour-coded boxes: Commission Amount = (Commission Rate ÷ 100) × Transaction Value; and Percentage Commission = (Commission Amount ÷ Transaction Value) × 100. Learners then individually solve three graded problems rooted in the phenomenon: (Easy) An Airtel Money agent in Kisumu charges KSh 25 on a KSh 1,000 transfer. What is the percentage commission? (Medium) Amina's M-PESA agent charges a 3% commission. Amina sends KSh	Write formulas on the board in two colours — orange for Commission Amount formula, blue for Percentage Commission formula. Say: 'These two formulas are mirror images of each other — just like Lessons 2 and 3 showed us that Discount and Percentage Discount are mirrors. Copy them into your books and draw a box around each.' Model the Easy problem aloud, narrating every step: 'Commission Amount = KSh 25. Transaction Value = KSh 1,000. So Percentage Commission = (25 ÷ 1,000) × 100 = 2.5%.' Say: 'Now solve the Medium and Challenge problems alone — no talking — you have 5 minutes.' [Circulate, crouch to eye level with	Worked Example → Gradual Release → Peer Marking (I Do / You Do / We Check): this three-step structure follows the NGSS practice of 'Using Mathematics and Computational Thinking.' The teacher models one problem explicitly, learners attempt two independently, then peer-marking creates a low-stakes accountability loop. Explicit connection back to the phenomenon at the end of the Explain phase ensures that procedural fluency is always anchored to meaningful context.	Collect 6 randomly selected books after the Check-and-Correct protocol and scan for: (1) correct formula application (are learners dividing by 100 rather than multiplying by 100?); (2) correct identification of the transaction value as the denominator in the percentage commission formula; (3) correct subtraction to find net amount received. Any learner who answers the Challenge problem correctly (percentage commission = 5%) demonstrates mastery and can be designated as a peer-tutor for the next lesson's practice work.

	800. How much does her aunt in Uganda receive? (Challenge) A real-estate broker in Nairobi sells a plot for KSh 850,000 and earns a commission of KSh 42,500. What is the broker's percentage commission? After individual working time, learners compare answers with a partner using a 'Check-and-Correct' protocol: each partner marks the other's work and identifies any step where a mistake occurred.	struggling learners, and ask: 'What do you already know from this problem? What is the transaction value?'] After 5 minutes: 'Swap books with your partner. Mark their work — put a tick if correct, a question mark if you are unsure.' Cold-call 3 learners for the Medium problem: 'Kamau, what answer did you get?' 'Fatuma, did you get the same? Where did your methods differ?' Explicitly connect back to the phenomenon: 'Amina is about to send money to Uganda. If the agent charges 3%, and Amina sends KSh 800, her aunt receives only KSh 776 — not KSh 800. Does that change Amina's plan?' [WAIT 15 seconds for quiet thinking before cold-calling.]		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Genetics vocabulary] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Repaying Money (PLIX)] Source: CK-12 🔍 [Search ARES for similar readings]	Learners return to the class Driving Question Board — a large poster on the wall that has been building since Lesson 1 with sticky notes and evidence cards. Each learner writes one new sticky note in the following format: 'We now know that [blank] because [blank] — and this means for Amina that [blank].' Examples might include: 'We now know that commissions are paid to intermediaries (like M-PESA agents) because they provide a service — and this means for Amina that sending KSh 800 will only deliver KSh 776 to her aunt in Uganda.' Learners post their sticky notes under the column 'New Evidence — Lesson 4: Commission.' They also revisit the sticky notes from Lessons 2 and 3 (about discounts) and identify one connection: 'Both discounts and commissions are expressed as percentages — but they affect different people in the transaction.' This connection is written on a	Point to the DQB and say: 'Look at what we have built since Lesson 1. Amina's story is getting more complicated — and more mathematical.' Read aloud one sticky note from Lesson 3 about discounts. Say: 'Today we added a new character to Amina's story — who is it?' [Cold-call 2 learners — expected answer: the mobile-money agent.] Say: 'Write your sticky note using the sentence frame on the board. You have 3 minutes.' [Display sentence frame: 'We now know that ___ because ___ — and this means for Amina that ___.'] Circulate and read over shoulders — prompt any learner who writes a vague note: 'Can you be more specific — use a number from today's calculations.' After posting, hold up a green bridge card and say: 'Who can tell me one thing that discounts and commissions have in common, and one thing that is different?' [WAIT 12 seconds.] Cold-call 3	Driving Question Board (Anchored Claim Construction): the DQB is a living document that externalises the class's growing understanding of the phenomenon. By requiring learners to write in the sentence frame 'We now know… because… and this means for Amina…', the teacher ensures that each entry is a claim backed by evidence and connected to the storyline — mirroring the NGSS practice of 'Constructing Explanations.' The green bridge card strategy makes cross-lesson connections visible and cumulative.	Read all posted sticky notes after class. Look for three quality indicators: (1) Is the claim specific (mentions a number or a named agent/context)? (2) Is the evidence cited (references the role-play data or formula result)? (3) Is the connection to Amina explicit? Sticky notes that meet all three criteria indicate a learner who has achieved the lesson's knowledge and skills outcomes. Sticky notes that are vague or only descriptive ('commissions exist') indicate learners who need reinforcement in the next lesson.

	green 'bridge card' and posted between the Lesson 2/3 column and the Lesson 4 column.	learners. Write their responses on the bridge card collaboratively.		
Model Building Phase 📹 **VIDEO:** Genetics vocabulary Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Repaying Money (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their 'Amina's Money Model' — a cumulative diagram started in Lesson 1 that maps where Amina's KSh 4,500 flows. In Lesson 1, they drew the three destinations (club trip, smartphone, Uganda transfer). In Lessons 2 and 3, they annotated the smartphone branch with discount calculations. Today, they annotate the Uganda transfer branch with commission calculations. Using pencil so they can revise, learners draw a new branch from 'Uganda Transfer' labelled 'M-PESA Agent Commission' and calculate: if Amina sends KSh 800 at a 3% commission rate, the agent earns KSh 24, and her aunt receives KSh 776. They write the formula used and the result next to the branch. Learners then answer a model revision question in their books: 'If Amina wants her aunt to receive exactly KSh 800, how much must Amina give the agent?' (Answer: KSh 800 ÷ 0.97 ≈ KSh 824.74, so approximately KSh 825.) This reverse-calculation step previews the algebraic thinking they will need in later lessons.	Say: 'Take out your Amina Money Model from Lesson 1. Today we are adding a new branch — the commission branch.' Display a model on the board and annotate the Uganda Transfer branch live as learners annotate their own. Say: 'Write the formula, substitute the numbers, and box your final answer — how much does Amina's aunt receive?' [Allow 3 minutes.] After annotation: 'Now here is a challenge that flips the problem. If Amina wants her aunt to receive exactly KSh 800 — not KSh 776 — how much must she give the agent?' [WAIT 15 seconds.] Say: 'Discuss with your partner for 1 minute, then write your answer.' Cold-call 2 pairs. If no pair gets the reverse calculation, guide: 'What if we call the amount Amina sends X? Then the agent takes 3% of X, so the aunt receives 97% of X. We want 97% of X to equal KSh 800. Can you solve that?' Do not solve it fully — leave it as a productive challenge. Say: 'We will come back to this type of problem in a future lesson. For now, add a question mark to that branch on your model — it is evidence of something we still need to learn.' Connect explicitly to the driving question: 'Amina's KSh 4,500 is shrinking — first the smartphone cost more than she thought because of the marked price, then the discount only saved part of the gap, and now the agent is taking a commission. How can she make smart decisions with what she has left?'	Cumulative Model Revision (Annotated Flow Diagram): the Amina Money Model is a boundary object that spans all eight lessons. Each lesson adds a new annotated branch, making the model progressively more complete. This mirrors the NGSS practice of 'Developing and Using Models' — the model is not decorative; it is a working tool for reasoning about the phenomenon. The productive challenge (reverse calculation) introduces cognitive disequilibrium intentionally, motivating future learning and signalling that the model is not yet complete.	Collect 5 models at the end of the lesson (rotate collection so all learners are reviewed across the sequence). Check: (1) Is the commission branch correctly labelled with the formula, substituted values, and result? (2) Is the arithmetic for the commission amount and recipient amount correct (KSh 24 and KSh 776 respectively)? (3) Did the learner attempt the reverse calculation — even an incorrect attempt shows engagement with the challenge? Models with all three elements complete and correct demonstrate full mastery of today's SLO.

D. TEACHER REFLECTION
1. During the role-play, did all pairs correctly identify the commission amount by physically counting out the dummy currency, or did some pairs skip the physical counting and go straight to the formula? If so, what does this tell me about those learners' relationship with concrete versus abstract representations, and how should I adjust the next practice activity? 2. When I cold-called learners during the Explain phase to connect the commission calculation back to Amina's phenomenon, did learners spontaneously use the word 'commission' confidently, or did they revert to 'fee' or 'charge'? What does this suggest about how well the formal vocabulary has been internalised? 3. How many sticky notes on the DQB met all three quality criteria (specific claim, cited evidence, connection to Amina)? Which learners' notes were vague, and what support do I need to put in place before Lesson 5? 4. The reverse-calculation challenge (how much must Amina send so her aunt receives exactly KSh 800?) was left deliberately open. How many learners attempted it, and did any reach the correct answer of approximately KSh 825? Should I open Lesson 5 by revisiting this challenge before moving to profit and loss? 5. Did the role-play remain orderly and mathematically focused, or did pairs finish at very different speeds? If there was a significant pacing gap, should I prepare an extension transaction card (e.g., a Nairobi insurance broker earning 8% commission on a KSh 120,000 annual premium) for fast finishers in future iterations of this lesson?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed — through a role-play at a simulated Kericho mobile-money kiosk — that an agent retains a fixed percentage of every transaction value as a commission. By counting dummy currency and recording data in a table, they saw that the commission amount grows proportionally with the transaction value, while the commission rate (percentage) stays constant at 3% across all three transactions.
What did I learn?	Commission is an amount paid to an agent or intermediary for facilitating a financial transaction; it is calculated as Commission = (Commission Rate ÷ 100) × Transaction Value. Percentage Commission = (Commission Amount ÷ Transaction Value) × 100. Both formulas are mirrors of each other, just as Discount and Percentage Discount were mirrors in Lessons 2 and 3. The recipient of a money transfer always receives less than the amount sent, because the agent deducts a commission.
How does this explain the phenomenon?	This explains a key part of Amina's phenomenon: when she sends money to her aunt in Uganda via a mobile-money agent, her aunt does not receive the full amount. If Amina sends KSh 800 and the agent charges 3% commission, the agent earns KSh 24 and Amina's aunt receives only KSh 776. For Amina to send enough money to cover her aunt's needs AND the agent's commission, she must calculate the gross amount to send — a reverse-commission problem that will be explored further in the sequence. The 'same KSh 4,500 producing different outcomes' in the driving question now has a new mathematical explanation: commissions are hidden costs that reduce the value of every transfer.

LESSON 5 (40 minutes): Profit and Loss (Part 1 — Concepts & Calculation) — From Wakulima Market to Amina's Stall	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners use buying-and-selling role-plays set in Wakulima Market and Kondele stalls to determine cost price, selling price, profit or loss, and percentage profit or loss — equipping Amina (and themselves) to judge whether a trade deal is truly good.
Knowledge	– Define cost price (CP) as the price paid to acquire goods and selling price (SP) as the price received when goods are sold. – State that Profit = SP − CP (when SP > CP) and Loss = CP − SP (when CP > SP). – Calculate Percentage Profit = (Profit ÷ CP) × 100 and Percentage Loss = (Loss ÷ CP) × 100. – Recognise that percentage profit/loss is always expressed relative to the cost price, not the selling price.
Skills	– Apply the profit and loss formulas to real-life buying-and-selling scenarios involving local goods (githeri, beans, second-hand clothes, mandazi). – Compare percentage profit or loss across different transactions to identify the better deal. – Present buying-and-selling calculations clearly to peers using correct mathematical language.
Attitudes	– Appreciate that honest and accurate calculation of profit and loss is essential for fair and sustainable trade. – Develop financial responsibility and consumer awareness when evaluating whether a transaction is profitable or results in a loss.
Key Inquiry Question	Why are discounts and commissions necessary in trade — and how do profit and loss calculations help every trader and buyer make fair, informed decisions?
Purpose in Storyline	Amina has already planned her budget (L1), calculated the discount on the smartphone (L2–L3), and understood the commission her mobile-money agent charges (L4). Now she must judge whether her mandazi-selling business actually makes a profit — and whether the second-hand smartphone stall at Kondele is giving her a fair deal or quietly making a large profit at her expense. This lesson provides the mathematical tools to answer those questions.
Safety Notes	No specific hazards. Ensure role-play scenarios are conducted respectfully; remind learners not to use real money during activities.

B. LESSON OVERVIEW

In this fifth lesson of the evidence-gathering sequence, learners shift focus from what buyers pay (discounts) and what agents earn (commissions) to what traders gain or lose. The anchoring context is a maize trader at Nairobi's Wakulima Market who buys a 90 kg bag of maize at KSh 2,700 and sells it in smaller portions — a scenario that mirrors Amina's own mandazi business in Kisumu. By working through this familiar context, learners construct the four core formulas of profit and loss (Profit, Loss, Percentage Profit, Percentage Loss) from first principles rather than memorising them in the abstract.

The lesson is structured around a structured role-play marketplace activity in which learner groups trade fictional quantities of githeri, beans, and second-hand clothes using price cards. Each group alternates between playing the trader and the buyer, records their CP and SP on a trading ledger sheet, then calculates profit or loss and its percentage. Groups present their results to the class, creating a shared data set that reveals how the same goods can produce very different percentage outcomes depending on how well the trader buys and at what price they sell. This connects directly back to Amina: the Kondele electronics stall that sold her a smartphone at "20% off" the marked price was still making a significant profit — learners calculate exactly how much.

By the end of the lesson, learners can move fluently between the four formulas, express profit and loss as percentages of cost price, and articulate why percentage profit/loss (rather than absolute profit/loss) is the fairer measure of a transaction's success. The Driving Question Board is updated to show that the same KSh 4,500 produces different outcomes not just because of discounts and commissions, but also because of the profit margin embedded in every price Amina encounters.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Modeling with one-step equations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Avoiding Soil Loss (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a 'Trader's Card' (a half-page slip) showing a scenario: 'Mama Akinyi bought a 2 kg packet of beans at Wakulima Market for KSh 120, then sold it at her stall for KSh 150. Did she make a profit or a loss? By how much? How would you express that as a percentage?' Learners write their prediction individually in 2 minutes — including a guess at the percentage — before any instruction. They then share predictions with a partner and note any disagreements on their card.	Distribute Trader's Cards as learners settle. Say: 'Before we do any mathematics together, I want to see what you already think. Read your card, write your prediction — profit or loss, the amount, and your best guess at the percentage. You have two minutes. Work alone first.' Set a visible timer. After 2 minutes, say: 'Now turn to your partner — 30 seconds each — share your prediction and listen to theirs.' After sharing, cold-call 3 pairs (choose one confident, one mid-range, one hesitant learner): 'Baraka, what did you and your partner predict?' WAIT 12 seconds after each response before affirming or probing. Record the three different percentage guesses on the board under the heading 'Our Predictions' — do NOT correct them yet. Say: 'We will come back to these predictions at the end of the Observe phase to see who was closest — and more importantly, why.'	Elicit-then-Refine Prediction: by committing a guess to paper before instruction, learners activate prior knowledge (from lessons on discounts and commissions) and create a personal stake in finding the correct answer. Disagreements between partners surface misconceptions — especially the common error of calculating percentage profit relative to selling price rather than cost price — which the teacher can target during the Explain phase.	Teacher scans Trader's Cards as they circulate during partner sharing. Look for: (1) learners who correctly identify profit of KSh 30 — they have the subtraction right; (2) learners who divide 30 by 150 (SP) instead of 120 (CP) for the percentage — flag this as the key misconception to address; (3) learners who write 'I don't know how to get the percentage' — prioritise these learners for targeted questioning during Observe.
Observe Phase 📹 **VIDEO:** Modeling with one-step equations Source: Khan Academy (English - US curriculum) 🔍 Search ARES	Learners participate in a structured 'Wakulima Marketplace' role-play in groups of four. Each group receives: a set of 6 laminated 'goods cards' (githeri 1 kg, dry beans 2 kg, second-hand shirt, mandazi dozen, maize flour 2 kg, second-hand school bag), a 'Cost Price' strip (the price the 'trader'	Before releasing groups say: 'You are traders at Wakulima Market — one of Nairobi's biggest open-air markets. Your job is simple: look at what you paid for each item (Cost Price) and what you sold it for (Selling Price), then record whether you made a profit, a loss, or broke even, and calculate the	Structured Role-Play with Ledger Recording: the physical trading simulation grounds abstract formulas in concrete action — learners experience the asymmetry between buying and selling price before they name it mathematically. The ledger format (CP | SP | Profit/Loss | Amount)	Teacher examines Trading Ledger worksheets during circulation. Specific indicators: (1) Is CP correctly identified as the wholesale/buying price and SP as the selling price — or are learners confusing the two? (2) Is the subtraction performed in the correct direction (SP − CP for

for similar videos 📖 **HTML:** Avoiding Soil Loss (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	paid at wholesale), and a 'Selling Price' strip (the price they charge customers). Traders record CP, SP, and calculate Profit or Loss in a simple Trading Ledger table on their worksheet. After 8 minutes of trading, groups rotate roles (traders become buyers) and a new set of price strips is used for a second round with different margins — one item deliberately sells below cost price to introduce loss. Each group records at least 4 completed ledger rows.	actual amount.' Circulate continuously. At the 4-minute mark, pause the room: 'Freeze — quick check. Hold up your worksheet. Everyone should have at least two rows filled in the CP and SP columns. If you have less than two, raise your hand.' Support struggling groups immediately. At the 7-minute mark, prompt: 'In 60 seconds, look at all your rows — find the one item where the trader actually lost money. Put a red star next to it.' Cold-call 2 groups to share which item caused a loss and why (e.g., 'We sold the second-hand shirt for KSh 200 but paid KSh 250'). WAIT 10 seconds after each group shares. After role-play, direct groups to return to Mama Akinyi's beans scenario from the Predict phase and formally calculate the profit and percentage: 'Now use what you just did in the market to go back to Mama Akinyi. What is her profit? What percentage is that of what she paid?'	creates a visual scaffold that makes the subtraction direction (SP − CP) feel intuitive rather than arbitrary. The deliberate inclusion of a loss scenario prevents learners from assuming trade always yields profit.	profit, CP − SP for loss)? (3) Can learners identify the loss item unprompted? (4) On returning to Mama Akinyi: does the learner now divide by CP (120) rather than SP (150)? Flag any group still dividing by SP for explicit correction in the Explain phase.
Explain Phase 📹 **VIDEO:** Modeling with one-step equations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Avoiding Soil Loss (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups present one trading scenario each (2 minutes per group, teacher selects 3 groups to present). Each presenting group writes their CP, SP, Profit/Loss, and percentage on the board. The class checks each group's arithmetic and votes (thumbs up / thumbs sideways / thumbs down) on whether the percentage was calculated correctly. After the three presentations, learners individually complete a 'Formula Consolidation Box' in their exercise books: they write the four formulas in their own words, then solve two structured problems — (a) a Kondele second-hand clothes trader who bought a dress for KSh 400 and sold it for	Manage presentations: 'Group 1 — come write your scenario on the board. Tell us: what was your CP, what was your SP, and what formula did you use?' After each presentation, ask the class: 'Thumbs up if you agree with the percentage calculation; thumbs sideways if you are unsure; thumbs down if you think it is wrong.' WAIT 8 seconds for thumbs to settle, then cold-call one thumbs-down learner: 'Chiamaka, you have thumbs down — what do you think went wrong?' WAIT 12 seconds. If the key misconception (dividing by SP) appears, address it explicitly: 'This is the most common mistake in profit and loss	Peer Critique with Formula Anchoring: the public presentation + thumbs voting creates a low-stakes error-correction culture where misconceptions are named collectively rather than attributed to one learner. Writing formulas in their own words (Formula Consolidation Box) promotes deep encoding. The Kondele smartphone extension problem re-anchors the new formulas directly to Amina's phenomenon, showing learners that discounts (L2–L3) and profit (L5) are two sides of the same transaction — the seller's profit is embedded in the pre-discount price.	During presentations: count how many groups used CP (not SP) as the denominator — this is the primary indicator of conceptual understanding. During Formula Consolidation Box: circulate and look for (1) correct formula statements, (2) correct identification of which number is profit and which is CP in each problem. For the Kondele extension: correct answer is SP after discount = KSh 5,000 × 0.80 = KSh 4,000; Profit = KSh 4,000 − KSh 3,200 = KSh 800; Percentage Profit = (800 ÷ 3,200) × 100 = 25%. Learners who arrive at 25% have fully integrated discounts and profit — note these for extension

	KSh 550, and (b) a githeri seller in Kisumu who bought ingredients for KSh 180 and sold a portion for KSh 150 — calculating profit/loss and percentage for each.	— the percentage is always out of what you PAID, not what you RECEIVED. The cost price is the base. Let me show you why that makes sense: if I pay KSh 100 and sell for KSh 200, I doubled my money — 100% profit. If I divided by 200 I would only get 50% — that doesn't match the story.' Write on the board: 'Percentage Profit = (Profit ÷ CP) × 100. Percentage Loss = (Loss ÷ CP) × 100. Always ÷ CP.' After consolidation box, cold-call 2 learners for problem (a) and 2 for problem (b). WAIT 10 seconds after each answer. Connect back to phenomenon: 'Now — what percentage profit is the Kondele stall making on the smartphone it sold Amina? The stall bought it for KSh 3,200 and marked it at KSh 5,000 before the 20% discount. What is the stall's percentage profit on the discounted sale price?' Give 2 minutes for learners to calculate individually.		tasks in Lesson 6.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Modeling with one-step equations](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Avoiding Soil Loss (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	Each learner writes one sticky note (or a strip on shared paper) responding to the prompt: 'What does today's lesson about profit and loss help us understand about Amina's situation — or what new question does it raise?' Learners post their notes on the class DQB under one of three columns: WHAT WE NOW UNDERSTAND | WHAT WE ARE STILL WONDERING | HOW THIS CONNECTS TO AMINA. The class does a 90-second silent gallery walk of the board before the teacher facilitates a 2-minute whole-class synthesis.	Distribute sticky notes. Say: 'Think about everything we have done today — the Wakulima Market role-play, Mama Akinyi's beans, and the Kondele smartphone calculation. Write one thing that now makes more sense about Amina's situation, OR one question you still have. You have 90 seconds.' After posting, say: 'Stand up — take 90 seconds to read what your classmates wrote. Don't talk yet, just read.' After gallery walk, facilitate synthesis by reading two notes aloud — choose one 'understanding' note and one 'wondering' note. Ask: 'Does anyone want to respond to this wondering?' WAIT 12 seconds. Explicitly update the DQB by	Three-Column DQB Sorting: organising responses into 'Understand / Wonder / Connect to Amina' prevents the DQB from becoming a list of disconnected questions. The sorting act itself is a metacognitive move — learners must decide whether they feel settled or unsettled about an idea before posting. The gallery walk creates social validation: learners see that peers share their confusions, reducing anxiety about not yet understanding everything.	Teacher reads all sticky notes after the lesson. Look for: (1) notes that correctly name the relationship between discount and profit (e.g., 'Even after the 20% discount, the stall still profits'); (2) notes that reveal persistent confusion about the direction of subtraction; (3) notes that raise forward-looking questions (e.g., 'What if Amina sells some mandazi at a loss to attract customers?') — these indicate readiness for Lesson 6's extended scenarios. Count the ratio of 'understand' to 'wonder' notes as a class-level confidence barometer.

		adding a teacher-written card: 'LESSON 5 UNLOCK: Every price Amina pays has a profit margin built into it — discounts reduce what she pays, but the seller may still profit. Commissions (L4) reduce what the agent passes on. Profit/loss tells us who really benefits from a transaction.' Point to previous DQB cards from L1–L4 and say: 'We are getting closer to being able to fully answer Amina's question — next lesson we go deeper into profit and loss calculations with more complex scenarios.'		
Model Building Phase 📹 **VIDEO:** [Modeling with one-step equations] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Avoiding Soil Loss (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners return to their cumulative 'Amina's Financial Journey' model — a diagram or annotated flowchart they have been building since Lesson 1. In Lesson 1 they drew income and expenditure flows; in L2–L3 they added discount arrows; in L4 they annotated the mobile-money transfer with a commission deduction. Today, learners add a new layer: they annotate the mandazi-selling activity with CP (ingredients cost), SP (selling price per batch), and calculate and label the profit and percentage profit Amina earns per batch. They also add a note to the Kondele smartphone transaction: the stall's profit margin even after giving Amina a 20% discount. Learners use a different colour (red for loss risk, green for profit) to code the direction of each money flow in their model.	Display the cumulative model from a previous lesson on the board (or a class reference copy). Say: 'Open your Amina Financial Journey model — find the section where Amina sells mandazi. We now know enough to add profit and loss to this part of her story. Using today's formulas, annotate her mandazi sales: she buys ingredients for KSh 300 per batch and sells the batch of mandazi for KSh 420. Add the Profit, Percentage Profit, and label them clearly. Then go to the smartphone transaction — add the stall's percentage profit in a thought bubble next to the stall.' Give 4 minutes for individual model revision. Circulate and ask targeted questions: 'Why did you put profit in green here?' WAIT 8 seconds. 'What would happen to Amina's model if she had to reduce the mandazi selling price to KSh 280 — would that section turn red?' WAIT 10 seconds. With 1 minute remaining, cold-call 2 learners to share one addition they made to their model and why. Close by saying: 'Your model now	Cumulative Annotated Model Revision: the colour-coded annotation (green = profit flow, red = loss risk) makes the directional nature of financial flows visible and persistent. By returning to the same artefact across all eight lessons, learners build a coherent, growing representation of Amina's financial world rather than a set of isolated calculations. Each revision act requires learners to integrate new knowledge with prior knowledge — the hallmark of deep conceptual learning.	Collect or photograph models at the end of the lesson. Success indicators: (1) Mandazi section shows CP = KSh 300, SP = KSh 420, Profit = KSh 120, Percentage Profit = (120 ÷ 300) × 100 = 40% — correctly labelled and colour-coded green; (2) Kondele smartphone section shows the stall's 25% profit margin annotated (from the Explain phase calculation); (3) Overall model shows logical flow connections between all annotated sections (budget → discount → commission → profit/loss). Models missing the percentage calculation or using SP as the denominator for percentage should be flagged for revision at the start of Lesson 6.

		shows budgets, discounts, commissions, AND profit and loss. In Lesson 6 we will push further — what happens when Amina faces more complex profit and loss situations involving multiple items and changing prices?'		

D. TEACHER REFLECTION

1. During the Predict phase, which misconception was most common — confusing profit with revenue, or dividing by SP instead of CP for percentage? How did you address it, and did learners who held that misconception correct it by the end of the Explain phase?
2. Was the Wakulima Marketplace role-play active enough to generate genuine engagement, or did some groups become passive? What adjustment (e.g., assigning specific trader/buyer roles with accountability tasks) would improve participation next time?
3. The Kondele smartphone extension problem required learners to chain together L2 (discount) and L5 (profit) knowledge. How many learners completed it correctly without prompting? What does this tell you about the depth of integration across the lesson sequence?
4. Reviewing the DQB sticky notes: what is the ratio of 'we understand' to 'we are still wondering' notes? Are the 'wondering' notes productive (pointing toward L6 content) or confused (suggesting gaps that need re-teaching)?
5. Did the colour-coded model revision reveal any learners whose mental model of Amina's financial journey is still fragmented (i.e., they treat each lesson as a separate topic rather than a connected story)? How will you support those learners' integration before Lesson 6?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners role-played as traders at Wakulima Market, buying and selling githeri, beans, second-hand clothes, and mandazi using price cards, and recorded cost prices and selling prices in a trading ledger. They observed that some transactions yielded profit while others resulted in a loss, and that the size of the margin varied widely between goods.
What did I learn?	Profit = SP − CP (when SP > CP); Loss = CP − SP (when CP > SP); Percentage Profit = (Profit ÷ CP) × 100; Percentage Loss = (Loss ÷ CP) × 100. The percentage is always calculated relative to the cost price. The Kondele electronics stall earns a 25% profit on the discounted smartphone even after giving Amina a 20% discount.
How does this explain the phenomenon?	Every price Amina encounters in the market — including the 'discounted' smartphone and the mandazi she sells — has a profit margin embedded in it. Understanding profit and loss mathematically allows Amina (and any Kenyan trader or consumer) to see beyond the surface price and evaluate whether a transaction is truly fair and financially sound.

LESSON 6 (40 minutes): Profit and Loss (Part 2 — Application & Problem Solving) — From Nyama Choma to Amina's Mandazi Stall	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners apply percentage profit and loss calculations to multi-step, real-life Kenyan business scenarios, consolidating Part 1 skills and reconnecting to Amina's mandazi-selling enterprise.
Knowledge	– Percentage Profit = (Profit ÷ Cost Price) × 100 and Percentage Loss = (Loss ÷ Cost Price) × 100 – Selling Price can be derived from Cost Price and a given percentage profit or loss: SP = CP × (1 + %Profit ÷ 100) or SP = CP × (1 − %Loss ÷ 100) – Multi-step profit/loss problems may involve overhead costs (e.g., transport, packaging) that must be added to the cost price before calculating profit or loss
Skills	– Set up and solve multi-step profit and loss problems systematically, identifying CP, SP, overhead costs, profit, and loss in each scenario – Interpret percentage profit and loss results to make and justify real-world financial decisions
Attitudes	– Appreciate that mathematical reasoning about profit and loss helps small-scale traders — like Amina and market vendors across Kenya — make fairer, smarter financial decisions – Develop persistence and accuracy when working through multi-step commercial problems
Key Inquiry Question	Why do we need a budget, and how do profit and loss calculations help Kenyan traders make smart money decisions?
Purpose in Storyline	Amina earned KSh 4,500 from selling mandazi. This lesson deepens learners' understanding of HOW she made that money — by calculating whether her mandazi business was genuinely profitable after accounting for ingredient costs and overhead — and connects profit/loss skills to Amina's decision to contribute to the club trip, buy the discounted phone, and send money to Uganda.
Safety Notes	No specific hazards. Ensure group work is conducted respectfully; remind learners that financial scenarios involve real dignity — avoid mocking peers' calculation errors.

B. LESSON OVERVIEW

This lesson is the sixth in the eight-lesson evidence-gathering sequence anchored on Amina's KSh 4,500 dilemma. Having established the core formulas for profit, loss, and their percentages in Lesson 5, learners now move into multi-step application and problem-solving. The lesson uses two richly contextualised Kenyan scenarios — a small nyama choma restaurant in Kericho and Amina's own mandazi stall at school — to give learners structured practice in identifying cost price (including overheads), computing profit or loss, and expressing results as percentages. By the end, learners will be able to determine whether a business is truly profitable after all costs are accounted for, a skill that directly answers the driving question about smart financial decision-making.

The lesson follows the five-phase ARES framework. In the Predict phase, learners estimate whether a nyama choma seller "made money" before doing any calculation. In the Observe phase, learner pairs work through two structured scenario cards — one involving the nyama choma restaurant and one modelled on Amina's mandazi business — recording all costs, revenues, and intermediate calculations. The Explain phase brings the class together to unpack the role of overhead costs and to generalise the multi-step calculation process. The DQB phase allows learners to update the class's growing understanding of the driving question, and the Model Building phase asks each learner to revise their personal financial model of Amina's KSh 4,500 to include a realistic profit calculation from her mandazi sales.

Throughout the lesson, the teacher uses targeted cold-calling, structured wait time, and peer explanation to surface and address common misconceptions — particularly the error of calculating percentage profit on selling price rather than cost price, and the error of forgetting overhead costs. All contexts are explicitly Kenyan: ingredients priced in Kenya Shillings, locations named (Kericho, Kondele, Kisumu), and the anchor character Amina kept central to every phase.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Modeling with systems of inequalities Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Avoiding Soil Loss (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners read a short scenario on the board: 'Mama Wanjiku runs a nyama choma stall in Kericho. On Saturday she bought 5 kg of goat meat for KSh 1,500, paid KSh 200 for charcoal, and sold the cooked nyama choma for KSh 2,100. Did she make a profit or a loss? How much? Write your prediction and one sentence of reasoning in your exercise book before we do any calculation.' Learners work individually for 2 minutes, committing to a prediction in writing.	Write the scenario on the board or display it before class. Say: 'Read carefully — do not calculate yet. I want your gut feeling and one reason. You have 2 minutes, starting now.' [Set a visible 2-minute timer.] Circulate silently — do NOT answer questions. After 2 minutes, say: 'Hands down. I am going to cold-call five people for their prediction and reason. Listen carefully because you may be asked to agree or disagree.' Cold-call 5 learners by name (aim for gender balance across the room). After each response, give 10 seconds of WAIT TIME before moving to the next, asking: 'Does anyone predict differently — and why?' Do not confirm or deny any prediction. Close by saying: 'Hold that thought. We are going to find out together.'	Prediction with Reasoning Commitment — learners write a prediction before instruction to activate prior knowledge from Lesson 5 and create a cognitive hook. Writing the reasoning (not just the answer) forces learners to articulate what they already understand about cost price, selling price, and profit, making subsequent learning more memorable and correctable.	Teacher scans written predictions during circulation — look for: (1) learners who correctly identify that overhead costs (charcoal) must be added to the cost of meat before comparing with revenue — this shows transfer from Lesson 5; (2) learners who compare KSh 2,100 against KSh 1,500 alone, ignoring charcoal — this is the key misconception to address in the Explain phase. Note at least 2–3 names in each category for targeted cold-calling later.
Observe Phase 📹 **VIDEO:** Modeling with systems of inequalities Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Avoiding Soil	Learners work in pairs. Each pair receives two scenario cards (or copies them from the board). Scenario A — Nyama Choma Kericho: Mama Wanjiku bought 5 kg of goat meat at KSh 300/kg, paid KSh 200 for charcoal and KSh 100 for a paper bag and seasoning, then sold the cooked nyama choma for KSh 2,400. Calculate (i) total CP, (ii) profit or loss, (iii) percentage profit or loss. Scenario B — Amina's Mandazi	Distribute scenario cards or write both scenarios clearly on the board. Say: 'Work with your partner. For each scenario, write out every step — do not skip. Label your Cost Price, your Selling Price, your Profit or Loss, and your Percentage. I will be walking around and I may ask either partner to explain their working.' Circulate actively — spend no more than 90 seconds with any one pair. Use these specific	Structured Pair Problem-Solving with Scenario Cards — pairing learners distributes cognitive load while maintaining individual accountability (either partner may be called). Using two scenarios — one unfamiliar (nyama choma) and one tied to the anchor character (Amina's mandazi) — allows learners to both practise the skill in a new context and immediately connect it back to the driving question phenomenon,	Teacher looks for: (1) correct identification of ALL cost components including overheads — key indicator of conceptual understanding; (2) correct calculation of actual revenue in Scenario B (84 mandazi × KSh 10 = KSh 840, not 90 × 10); (3) correct denominator in percentage calculations (CP, not SP). Record which pairs struggle with overhead costs and which confuse SP/CP — use this data in the Explain phase

Loss (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Stall: Amina spent KSh 600 on flour, sugar, and fat, and KSh 150 on charcoal for her jiko. She made 90 mandazi and sold each for KSh 10. However, 6 mandazi were unsold. Calculate (i) actual revenue, (ii) total CP, (iii) profit or loss, (iv) percentage profit or loss. Pairs work through both scenarios systematically, showing all steps in their exercise books.	prompts when you see errors: If a pair forgets overhead costs, ask: 'What did Mama Wanjiku spend money on BESIDES the meat?' [Wait 10 seconds.] If a pair calculates % profit on SP instead of CP, say: 'Percentage profit is always calculated on — what price? Check your Lesson 5 notes.' [Wait 15 seconds.] For Scenario B, if a pair forgets the unsold mandazi, ask: 'Did Amina sell ALL 90 mandazi? Read again.' After 12 minutes, call time. Cold-call 3 different pairs to share their answer to Scenario A step-by-step, and 2 pairs for Scenario B. For each, ask the class: 'Do you agree? Thumbs up, thumbs sideways, or thumbs down.'	strengthening transfer.	to sequence cold-calls strategically.
Explain Phase 📹 **VIDEO:** Modeling with systems of inequalities Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Avoiding Soil Loss (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher leads a whole-class consolidation structured around the two scenarios. Learners follow along, correcting or confirming their own working. The teacher uses the board to build a 'Multi-Step Profit/Loss Framework' — a structured layout learners copy: Step 1: List ALL costs (CP + overheads) → Total Cost Price. Step 2: Calculate actual revenue (SP × units sold). Step 3: Compare — if Revenue > Total CP → Profit; if Revenue < Total CP → Loss. Step 4: Calculate percentage (always divide by Total CP × 100). Learners then answer one extension question independently: 'If Amina wants to make a 25% profit on her next mandazi batch with the same costs, what should she charge per mandazi (assuming she sells all 90)?' They write their answer and method before a partner checks it.	Begin by saying: 'Let us build this together. I heard some excellent thinking and some common mix-ups. We are going to sort both.' Write 'Multi-Step Profit/Loss Framework' as a heading on the board. Work through Scenario A publicly, pausing at each step: After Step 1 say: 'Total CP is KSh 1,500 + KSh 200 + KSh 100 = KSh 1,800. Who initially forgot one of these costs? Be honest — raise your hand.' [Acknowledge without shame: 'Good — that is exactly the most common mistake traders make in real markets.'] After Step 3 say: 'Revenue is KSh 2,400. Is that bigger or smaller than KSh 1,800? So, Mama Wanjiku made a — ?' [WAIT 10 seconds, cold-call 1 learner.] After Step 4 say: 'Percentage Profit = (600 ÷ 1,800) × 100 = 33.3%. Why do we divide by 1,800 and NOT by 2,400?' [WAIT 15 seconds, cold-call 2 different learners.] Repeat the framework for Scenario B, asking	Worked Example with Structured Framework + Error Spotting — presenting the multi-step framework explicitly while publicly acknowledging the most common error (forgetting overheads) normalises mistake-making and focuses attention on the structural logic of the calculation rather than rote memorisation. The extension question (working backwards from a desired profit percentage) deepens understanding by reversing the direction of reasoning.	Teacher assesses understanding by: (1) observing hand-raises when asking who forgot overhead costs — this gives a population-level indicator of the misconception's prevalence; (2) quality of explanations from the 2 cold-called learners explaining WHY percentage is calculated on CP not SP — look for the phrase 'profit is what you gained ON TOP OF what you spent'; (3) accuracy of the extension question — specifically whether learners correctly multiply CP by 1.25 (using the SP formula from Lesson 3) rather than adding 25% to per-unit revenue.

		learners to supply each number: 'What is Amina's Total CP? Revenue? Profit? Percentage?' [Cold-call 4 different learners, one per step.] For the extension question, give 3 minutes of silent individual work, then say: 'Turn to your partner and check each other's method — not just the answer, the method.' After 1 minute, cold-call 2 pairs to share their answer (expected: Total CP = KSh 750; SP needed for 25% profit = KSh 750 × 1.25 = KSh 937.50; per mandazi = KSh 937.50 ÷ 90 = KSh 10.42, so round up to KSh 11).		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Modeling with systems of inequalities Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Avoiding Soil Loss (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners spend 3 minutes individually writing on a sticky note (or a torn piece of paper): one thing they NOW understand about the driving question ('How can Amina make smart, fair, and informed money decisions?') that they did not fully understand before Lesson 6, AND one new question that today's lesson has raised for them. Learners post their notes on the class DQB under two columns: 'We now understand…' and 'We still wonder…'. The teacher reads 2–3 notes aloud and the class briefly discusses how today's lesson has moved understanding forward.	Say: 'You have 3 minutes. Write two things on your sticky note: first, one idea about Amina's money decisions that you now understand more deeply after today — be specific, use numbers or formulas if you can. Second, one new question today's lesson raised for you.' [Set a 3-minute timer. Circulate silently.] After 3 minutes: 'Post your note on the DQB — 'We now understand' on the left, 'We still wonder' on the right.' Once posted, read 2 'We now understand' notes aloud (choose ones that reference overheads or percentage profit on CP) and say: 'This learner has spotted something important — profit percentage is only meaningful if you include ALL your costs. Amina's 4,500 shillings came from REAL profit, not just cash received.' Then read 1 'We still wonder' note (ideally one that anticipates Lesson 7 — currency exchange) and say: 'We will get to that in our next lesson.' Point to the DQB and say: 'Notice how our board is filling up. Every lesson we	Driving Question Board Update with Two-Column Posting — the two-column structure (understanding vs. wondering) makes the progression of learning visible to learners over the eight-lesson sequence. Posting publicly creates accountability and community — learners see that the class as a whole is building understanding together, which mirrors how scientific and financial knowledge accumulates iteratively.	Teacher reads all posted notes after class. Look for: (1) notes that correctly articulate the overhead cost principle — these indicate conceptual transfer; (2) 'We still wonder' notes that reveal persistent misconceptions (e.g., 'why does the percentage change if you add overheads?') — flag these for brief revision at the start of Lesson 7; (3) notes that spontaneously connect to earlier lessons (discounts, commission) — these indicate learners are building an integrated mental model of the phenomenon.

		get closer to fully answering our driving question for Amina.'		
Model Building Phase 📹 **VIDEO:** [Modeling with systems of inequalities] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Avoiding Soil Loss (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners return to the personal 'Amina's Financial Model' diagram they began in Lesson 1 and have revised each lesson. Today they add a new annotated section labelled 'How Amina Earned KSh 4,500.' Using the Scenario B data from the Observe phase as a template, learners estimate or calculate: if Amina's typical mandazi batch costs KSh 750 (ingredients + charcoal) and she sells 84 mandazi at KSh 10 each, how many batches did she need to sell to accumulate KSh 4,500? They annotate their model diagram with the calculation, a small arrow showing 'profit per batch → savings → KSh 4,500', and a note connecting this to the club trip, phone purchase, and money transfer from the anchoring phenomenon.	Say: 'Open your Amina model diagram. We are going to make it more complete today. Add a new section at the top: How did Amina actually EARN her KSh 4,500? Use today's Scenario B numbers as your guide.' Give learners 4 minutes of quiet individual work. Circulate and prompt with: 'If she makes KSh 90 profit per batch, how many batches to reach KSh 4,500? Show that calculation on your diagram.' [WAIT — do not give the answer.] After 4 minutes, cold-call 1 learner: 'Tell us your calculation for the number of batches.' [Expected: KSh 4,500 ÷ KSh 90 = 50 batches.] Say: 'Does that feel realistic for a school term? How many school days are in a term? Could Amina sell mandazi most days?' [Allow 30 seconds of open discussion.] Close by saying: 'Your model now connects Amina's hard work — 50 batches of mandazi, each carefully priced — to the KSh 4,500 she is trying to spend wisely. Every decision she makes with that money — the discount, the commission, the exchange rate — starts HERE, with profit and loss.' Ask learners to keep their model safe: 'We will add the exchange rate in Lesson 7.'	Cumulative Model Revision — returning to the same diagram each lesson makes the growth of understanding tangible and prevents individual lessons from feeling isolated. Adding a quantitative back-story (number of batches) to the anchor phenomenon deepens empathy for Amina as a real economic actor and reinforces that commercial arithmetic is not abstract — it governs the savings of real Kenyan young people.	Teacher collects or photographs 4–5 model diagrams (rotate which learners each lesson) to check: (1) correct calculation of profit per batch (Revenue KSh 840 − CP KSh 750 = KSh 90 profit); (2) correct division to find number of batches (KSh 4,500 ÷ KSh 90 = 50); (3) meaningful annotation connecting the profit calculation to the three spending decisions in the phenomenon (club trip, phone, money transfer). Models that show all three connections indicate the learner is successfully integrating the lesson sequence into a coherent understanding of the driving question.

D. TEACHER REFLECTION

1. During the Observe phase, which misconception appeared more frequently — forgetting overhead costs or calculating percentage profit on selling price rather than cost price? How will you address the more prevalent one at the start of Lesson 7?

2. When learners posted on the DQB, did the 'We still wonder' notes reveal any gaps in understanding from earlier lessons (Lessons 1–5) that need revisiting, or were they genuinely anticipatory questions about upcoming content (exchange rates, currency conversion)?

3. Did all learners successfully connect Amina's KSh 4,500 savings to a realistic number of mandazi batches in the Model Building phase? If some produced implausible

numbers (e.g., 1 or 2 batches), what does that suggest about their grip on the relationship between unit profit, scale, and total savings? 4. Were gender and ability groups balanced in your cold-calling pattern across today's five phases? Review your mental notes — did the same 4–5 confident learners dominate responses, and if so, how will you structure cold-calling more equitably in Lesson 7? 5. The extension question asked learners to work BACKWARDS from a desired profit percentage to a required selling price — a reversal of the standard direction. How many learners attempted this confidently, and what does their success rate tell you about depth of conceptual understanding vs. procedural recall?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed two multi-step scenarios — Mama Wanjiku's nyama choma stall in Kericho and Amina's mandazi stall — in which total cost price included both direct ingredient costs and overhead costs (charcoal, packaging). They observed that ignoring overhead costs produces an inflated and misleading profit figure, and that in Amina's case not all units produced were sold, further affecting revenue.
What did I learn?	Percentage profit and loss must always be calculated on the TOTAL cost price, which includes all overheads, not just the purchase price of goods. The multi-step framework is: (1) Total CP = direct costs + overheads; (2) Actual Revenue = SP × units actually sold; (3) Profit or Loss = Revenue − Total CP; (4) % Profit or Loss = (Profit or Loss ÷ Total CP) × 100. A desired selling price can be reverse-engineered from a target profit percentage using SP = CP × (1 + %Profit ÷ 100).
How does this explain the phenomenon?	Amina's KSh 4,500 was not simply cash received — it represents genuine profit accumulated over approximately 50 mandazi batches after all ingredient and fuel costs were covered. This means every shilling of her KSh 4,500 required careful cost management. When she now faces a discount at the Kondele stall, a commission on the mobile-money transfer, and a currency exchange for her aunt in Uganda, she is making decisions with money she earned through exactly the kind of commercial arithmetic this lesson practised — confirming that profit and loss calculations are foundational to all of Amina's financial decisions.

LESSON 7 (40 minutes): Currency Conversion — Crossing Borders with Amina's Shillings	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners use simulated Central Bank of Kenya forex bureau exchange-rate tables to convert Kenya Shillings to Uganda Shillings, US Dollars, and Euros — and back again — while discovering why buying rates differ from selling rates, directly solving Amina's cross-border transfer problem.
Knowledge	– Define 'exchange rate', 'buying rate', and 'selling rate' as used by forex bureaux and mobile-money agents. – Explain why the buying rate is always lower than the selling rate (the spread is the bureau's commission/profit). – State the formula: Amount in foreign currency = Amount in KES ÷ Selling Rate (when buying foreign currency); Amount in KES = Amount in foreign currency × Buying Rate (when selling foreign currency back).
Skills	– Read and interpret a simulated CBK forex bureau exchange-rate table to extract the correct buying or selling rate for a given transaction. – Perform multi-step currency conversion calculations (KES → UGX, KES → USD, KES → EUR, and reverse) and apply them to real financial decisions like Amina's transfer to her aunt in Uganda. – Identify the hidden commission embedded in the buying-selling rate spread and link it to Lesson 4 knowledge of percentage commission.
Attitudes	– Demonstrate patriotism and financial citizenship by appreciating Kenya's role in the East African monetary community and the importance of fair forex rates. – Develop critical consumer awareness — questioning advertised rates before committing to a transaction — as a life skill for informed money management.
Key Inquiry Question	Why do discounts and commissions matter in trade — and how does the forex bureau's rate spread function as a form of commission?
Purpose in Storyline	Amina has budgeted, compared discounts, accounted for mobile-money commission, and calculated profit on her mandazi. Now she faces the final step: sending the remainder of her KSh 4,500 to her aunt in Uganda. This lesson unlocks currency conversion so learners can compute exactly how many Uganda Shillings Amina's aunt will receive, why the amount differs depending on which rate is used, and how the bureau profits from the spread — completing the full commercial arithmetic journey.
Safety Notes	No specific physical hazards. Remind learners to observe digital safety if accessing live exchange-rate websites on devices — they should use only the teacher-provided simulated CBK table during class, not live trading platforms.

B. LESSON OVERVIEW

This lesson is the penultimate evidence-gathering session in the Commercial Arithmetic 1 sequence and directly resolves one of the three strands of Amina's anchoring phenomenon: sending money across the Kenya–Uganda border. Learners receive a simulated Central Bank of Kenya (CBK) forex bureau exchange-rate table listing buying and selling rates for Uganda Shillings (UGX), US Dollars (USD), and Euros (EUR) against the Kenya Shilling (KES). Working in pairs, they use the table to answer increasingly complex conversion questions before arriving at Amina's specific problem: if she sends KSh 1,200 to her aunt in Kampala via a Kondele mobile-money agent, how many Uganda Shillings will the aunt receive, and what is the agent's implied commission from the rate spread?

The lesson is structured around a powerful conceptual link: in Lesson 4 learners discovered that commission is money paid to an agent for facilitating a transaction.

Today they discover that forex bureaux and mobile-money agents do not always charge a visible flat commission — instead, they embed their profit in the gap between the buying rate (the low price at which the bureau buys foreign currency from you) and the selling rate (the higher price at which it sells foreign currency to you). This "spread" is a form of percentage commission hiding in plain sight. By computing the spread as a percentage of the mid-rate, learners connect new knowledge to prior learning and deepen their understanding of financial literacy.

By the end of the lesson, learners will have solved Amina's Uganda transfer problem completely, updated the class Driving Question Board with new understanding about hidden costs in financial transactions, and revised their cumulative model of "smart money decisions" to include the principle: always check BOTH the buying and selling rate before any currency exchange.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Modeling with systems of inequalities Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Conversion of Decimals, Fractions, and Percent (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a sticky note. The teacher displays the prompt on the board: 'Amina has KSh 1,200 left after buying the smartphone and budgeting for the club trip. She wants to send it to her aunt in Kampala, Uganda. At the Kondele mobile-money agent, a sign reads: USD Buying 128.50 / Selling 129.80 | UGX Buying 0.034 / Selling 0.036. Predict: How many Uganda Shillings will Amina's aunt receive? Write your best guess AND one reason why you are unsure.' Learners write individually for 3 minutes, then do a think-pair-share with their desk partner.	Display the rate board and read it aloud slowly. Say: 'Look carefully — there are TWO numbers for every currency. Before I explain anything, I want your honest prediction. There is no wrong answer yet.' Start a 3-minute silent timer visibly on the board. After time, say: 'Turn to your partner — 30 seconds each, share your prediction and your uncertainty.' Then cold-call 3 pairs: 'Pair at the window — what did you predict and why are you unsure?' Allow a full 12-second WAIT TIME after each cold-call before accepting or redirecting. Record 3–4 different predictions on the board without evaluating them. Say: 'Hold these predictions — by the end of class we will know which is correct and WHY.'	Elicit-and-Record Prediction Wall — recording unevaluated student predictions publicly creates cognitive tension (Predict-Observe-Explain cycle). Seeing multiple conflicting answers on the board motivates learners to resolve the disagreement through the evidence phase, rather than passively receiving information.	Teacher scans sticky notes and listens to pair talk. Key indicators: (1) Does the learner attempt to multiply or divide by one of the two rates? (2) Does the learner notice there are two different rates and express confusion about which to use? Misconception to flag: learners who average the two rates or ignore one entirely — these learners need targeted support in the Explain phase.
Observe Phase 📹 **VIDEO:** Modeling with systems of inequalities Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Each pair receives the simulated 'CBK Forex Bureau Rate Table' (printed or displayed) showing buying and selling rates for UGX, USD, and EUR against KES, plus a worked reference box showing the general rules: 'To GET foreign currency, use the SELLING rate (divide KES by the selling rate). To GET back KES, use the BUYING	Distribute the rate table and worksheet. Say: 'The reference box tells you the two rules. Read it with your partner — you have 90 seconds before I take any questions.' After 90 seconds, cold-call one pair: 'Tell me in your own words: when does Amina use the selling rate?' Allow 10-second WAIT TIME. Circulate during the	Tiered Worked-Example Scaffolding — the reference rules box provides a minimally guided anchor before independent practice. Tiering tasks 1→4 from simple conversion to reverse conversion to Amina's real problem ensures every learner accesses at least Tasks 1–3 (the core SLO) while high-readiness	Teacher checks worksheets at circulation. Success indicator for Task 3: KSh 1,200 ÷ 0.036 = UGX 33,333 (rounded). Common error: multiplying instead of dividing (gives UGX 43.2 — flagrantly wrong, prompting self-doubt). For Task 4: UGX 38,000 × 0.034 = KSh 1,292. If a pair gets the same answer for buying and selling

📖 **HTML:** [Conversion of Decimals, Fractions, and Percent (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	rate (multiply foreign currency by the buying rate).' Pairs work through four graded tasks on a worksheet: Task 1 — convert KSh 5,000 to USD; Task 2 — convert USD 30 back to KES; Task 3 — convert KSh 1,200 to UGX (Amina's problem); Task 4 — Amina's aunt sends back UGX 38,000 to Kenya — how many KES does Amina receive? Learners record every step including which rate they chose and why.	12-minute working period. At each pair, ask: 'Which rate are you using here — and why?' rather than correcting errors directly. If a pair has the wrong rate, say: 'Who is giving up the KES in this transaction — Amina or the bureau?' to prompt self-correction. At the 10-minute mark, call attention: 'Pairs who have finished Task 4, calculate this: what is the difference in KES between buying and selling rates on 1 USD? We will need that number in the Explain phase.' Give 2 additional minutes for this extension.	pairs reach the extension that bridges to the commission concept. Recording 'which rate and why' forces metacognitive justification, not just calculation.	conversions, they have used the same rate twice — pause the pair and ask them to re-read the reference rules aloud.
Explain Phase 📹 **VIDEO:** [Modeling with systems of inequalities] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Conversion of Decimals, Fractions, and Percent (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The class reconvenes. The teacher reveals the correct answer to Task 3 (UGX 33,333) and compares it to the predictions recorded at the start of class. Learners then work through a guided discussion to discover the spread. The teacher writes on the board: 'UGX Buying 0.034 | Selling 0.036 | Difference = 0.002.' Learners calculate: (i) How many UGX does the bureau keep per KES 1 exchanged? (ii) Express this as a percentage of the mid-rate (0.035). This gives approximately 5.7% — a hidden commission. Learners then write in their exercise books: 'The rate spread is a form of commission. It is how the forex bureau makes money without showing a separate commission line.' A pair shares their Task 4 result and the class confirms: Amina gets back KSh 1,292 from UGX 38,000 — more than she sent — because the aunt topped up.	Return to the prediction board. Say: 'Look at what we predicted. Who was closest — and what made the difference between right and wrong?' Cold-call 2 learners (not the ones who answered in Predict). Allow 12-second WAIT TIME each. Then say: 'Now I want you to help me find the bureau's profit. On your worksheet, Task 3: Amina sent KSh 1,200 and the bureau used rate 0.036. If it had used the mid-rate of 0.035, Amina would have received how many more UGX?' Give 2-minute silent calculation time. Cold-call one learner for the answer (approximately UGX 952 more). Say: 'So the bureau kept the equivalent of UGX 952 — that is their commission hidden in the rate. We saw commission in Lesson 4 as a separate fee. Here it hides in the rate. How are they the same? How are they different? Discuss with your partner for 60 seconds.' Cold-call 2 pairs. Confirm: 'Both are payments to an agent for facilitating a transaction. The difference is transparency.' Summarise the two conversion	Concept-Bridge Discussion — explicitly connecting new knowledge (rate spread) to prior lesson knowledge (commission, Lesson 4) through a guided Socratic sequence. Naming the hidden commission aloud and writing 'The rate spread is a form of commission' in their own books converts declarative knowledge into transferable financial literacy. The comparison of 'transparent fee vs hidden spread' builds critical consumer thinking.	Teacher listens for: (1) Can learners articulate WHY the bureau's spread is a commission? (2) Can they calculate the UGX difference correctly? Exit check: teacher asks 3 learners at random to state — without notes — 'When do you divide, and when do you multiply?' Learners who cannot state the rule need a brief one-on-one consolidation before the DQB phase. Written indicator: the sentence in exercise books should mention 'agent', 'rate spread', and 'commission' in context.

		rules on the board in learners' own language from responses.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Modeling with systems of inequalities Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Conversion of Decimals, Fractions, and Percent (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board displayed at the front of the room. The board has columns: WHAT WE KNOW | WHAT WE WONDER | HOW IT CONNECTS TO AMINA. Each learner writes one new sticky note to add to the board based on today's lesson. Suggested prompts are on the board: 'Today I learned that…', 'I still wonder why…', 'This connects to Amina because…'. After placing their notes, the class reads three notes aloud selected by the teacher and discusses whether Amina's problem is now fully solved.	Give each learner one sticky note. Say: 'You have 2 minutes to write one thing — your choice of the three prompts on the board. Be specific: do not write 'I learned about exchange rates.' Write the actual number or rule you learned.' After placement, read 3 notes aloud (choose one from each column). For the 'CONNECTS TO AMINA' column, ask: 'Is Amina's transfer problem now completely solved? What information would she still need before walking into that Kondele agent's shop?' Allow 10-second WAIT TIME. Accept 2 responses. Say: 'She still needs to ask: does the agent also charge a flat fee on top of the spread? That is a real question for next lesson and for life.' Write this on the WONDER column of the DQB as a class-generated question.	Cumulative Evidence Board — the DQB tracks the class's growing understanding across all 8 lessons. Adding to it after each lesson externalises the learning progression and makes the storyline arc visible. Requiring specificity ('write the actual rule or number') prevents vague entries and raises the cognitive demand of reflection. The teacher's final 'what would Amina still need?' keeps productive uncertainty alive, sustaining inquiry momentum into Lesson 8.	Teacher reads all sticky notes after class. Indicators: (1) KNOW notes should reference buying rate, selling rate, division, or multiplication correctly. (2) WONDER notes should show genuine curiosity about forex systems (e.g. 'Why does the rate change daily?'). (3) CONNECT notes should mention Amina and at least one specific number or scenario. Sticky notes that are vague ('I learned maths') signal disengagement — teacher follows up individually at the start of Lesson 8.
Model Building Phase 📹 **VIDEO:** Modeling with systems of inequalities Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Conversion of Decimals, Fractions, and Percent (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners open their exercise books to their cumulative 'Smart Money Decisions Model' — a diagram or flowchart they have added to after every lesson in this sequence. Today they must add a new branch labelled 'Sending Money Across Borders' with three elements: (1) the two conversion rules (÷ selling rate to buy foreign currency; × buying rate to sell back), (2) the spread = hidden commission, and (3) Amina's final calculation showing how many UGX her aunt receives from KSh 1,200. Learners who finish early add a comparative note: 'If I sent the same KSh 1,200 to the USA, how many USD would arrive?' using the rate table.	Say: 'Open your cumulative model. You have added budgeting, discounts, commission, and profit-and-loss. Today you add the final transaction: cross-border money transfer. Draw or write your new branch — you have 4 minutes.' Circulate and specifically check that learners have written both rules (not just one) and have included the spread-as-commission insight. If a learner has only written the answer to Amina's problem without the rules, say: 'Good calculation — now add the rule so that next year when you forget the answer, you can re-derive it.' With 1 minute remaining, cold-call one learner to read their model entry aloud. Say: 'Notice how your model now covers	Cumulative Model Revision — requiring learners to physically add to a single running diagram across all lessons builds schema integration: new knowledge is not stored in isolation but connected to the existing mental model. Naming all six prior lessons aloud at the close activates retrieval practice and reinforces the coherence of the Commercial Arithmetic 1 sequence as a unified problem-solving framework rather than disconnected topics.	Teacher scans models during circulation. Success criteria: (1) Both conversion rules are present and correctly labelled (not just one direction). (2) The spread is described as a commission or 'hidden fee'. (3) Amina's UGX answer (approximately 33,333 UGX from KSh 1,200 at selling rate 0.036) is recorded. (4) The model connects visually or textually to at least one prior lesson element (e.g. an arrow from 'commission — Lesson 4' to 'rate spread = hidden commission — Lesson 7'). Models missing the spread-commission link are the primary remediation target for Lesson 8 warm-up.

		everything Amina needed: budget in Lesson 1, discount in Lessons 2 and 3, commission in Lesson 4, profit-loss in Lessons 5 and 6, and today, currency conversion. Next lesson we bring it all together to help Amina make her final decision. Look at how far your model has grown.'		

D. TEACHER REFLECTION

1. When learners made their initial predictions, how many used the correct rate (selling rate) spontaneously versus the buying rate or an average? What does this tell me about their prior understanding of the buyer/seller perspective in financial transactions — and how should I adjust the opening of Lesson 8?\n2. Did all learners successfully complete at least Tasks 1–3 on the worksheet within the time available? If not, which step caused the greatest bottleneck — reading the table, choosing the correct rate, or performing the arithmetic — and what targeted support will I provide?\n3. How effectively did learners make the connection between the rate spread and the commission concept from Lesson 4? Were they able to articulate the similarity (payment to an agent) and the difference (transparent vs. hidden) without heavy prompting, or did I need to provide the connection directly?\n4. Reviewing the sticky notes on the DQB: are the WONDER entries genuinely curious and mathematically specific, or are they superficial? What does the quality of wonder questions tell me about the depth of conceptual engagement in this class?\n5. Looking at the cumulative models: do learners' diagrams show genuine integration of all seven lessons' concepts, or are entries from earlier lessons disconnected from today's addition? If integration is weak, how will I structure the Lesson 8 synthesis activity to explicitly scaffold the connections?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners examined a simulated CBK forex bureau exchange-rate table showing buying and selling rates for UGX, USD, and EUR, and observed that every currency has TWO different rates — a lower buying rate and a higher selling rate — and that using the wrong rate produces a significantly different (and incorrect) conversion result.
What did I learn?	To convert KES into foreign currency, divide by the selling rate; to convert foreign currency back to KES, multiply by the buying rate. The difference between the buying and selling rate (the 'spread') is the forex bureau's embedded commission — a hidden profit that functions exactly like the percentage commission we studied in Lesson 4, but is concealed inside the rate rather than charged as a separate visible fee. Amina's KSh 1,200 sent at a selling rate of 0.036 yields approximately UGX 33,333 for her aunt in Kampala.
How does this explain the phenomenon?	This explains part of Amina's original puzzle: the same KSh 4,500 produces different outcomes depending on which financial transaction she uses it for. When sending money to Uganda, the exchange rate she sees on the sign is NOT the rate she actually transacts at — the bureau uses the selling rate (which is less favourable to Amina) to make its own profit. Just as the mobile-money agent in Lesson 4 charged a visible commission, the forex bureau charges an invisible one through the rate spread. Understanding both forms of commission is essential for Amina — and every Kenyan — to make smart, fair, and informed cross-border money decisions.

LESSON 8 (40 minutes): Synthesis, Model Revision & Final Explanation — Amina's Complete Financial Plan

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners synthesise all six commercial arithmetic learning outcomes by producing a complete, written financial plan for Amina — integrating budget, discount, commission, profit/loss, and currency conversion — and use peer review and DQB completion to confirm their final explanations of the anchoring phenomenon.
Knowledge	– A budget is a mathematical plan comparing income and expenditure that guides spending decisions in clubs, households, and businesses. – Discount (marked price − sale price) and percentage discount express the real saving on a purchase, and commission (rate × transaction value ÷ 100) is the cost of using a financial intermediary. – Profit = SP − CP and loss = CP − SP are always expressed as percentages of the total cost price, including overheads. – Currency conversion requires using the correct bank/agent rate: divide KES by the selling rate to obtain foreign currency, multiply foreign currency by the buying rate to return to KES. – All four operations interact simultaneously in real financial decisions, so errors in one calculation cascade into others.
Skills	– Construct a multi-step financial plan that correctly sequences budget preparation, discount calculation, commission deduction, profit/loss analysis, and currency conversion for a single scenario. – Evaluate a peer's financial plan against KICD assessment rubric criteria and provide specific, evidence-based written feedback. – Revise a cumulative explanatory model by comparing the initial (Lesson 1) sketch with the fully annotated final version and articulating what changed and why.
Attitudes	– Appreciate that mathematical precision in commercial arithmetic directly protects individuals — like Amina — from financial loss and exploitation in everyday Kenyan markets. – Value peer feedback as a collaborative tool for improving mathematical reasoning, reflecting the CBE competency of communication and collaboration. – Demonstrate patriotism and financial literacy by recognising how informed money management contributes to personal and national economic development.
Key Inquiry Question	How can Amina — and every Kenyan — use commercial arithmetic to make smart, fair, and informed money decisions in markets, clubs, and across borders?
Purpose in Storyline	This is the culminating lesson in which learners answer the driving question in full: they apply every concept from Lessons 1–7 to write Amina's complete financial plan, peer-review each other's work against rubric criteria, update the Driving Question Board with final agreed answers, and compare their initial naïve model (Lesson 1) with their fully annotated final model — making the growth of their mathematical understanding visible and explicit.
Safety Notes	No specific physical hazards. Remind learners to handle each other's written work respectfully during peer review and to give constructive, evidence-based feedback rather than personal criticism.

B. LESSON OVERVIEW

Lesson 8 is the final and integrative lesson of Sub-Strand 2.8. Over the preceding seven lessons learners have built understanding concept by concept — budgets

(Lesson 1), discounts (Lessons 2–3), commissions (Lesson 4), profit and loss (Lessons 5–6), and currency conversion (Lesson 7). In this lesson those seven strands converge. Learners return one final time to Amina, the Form 4 student from Kisumu who saved KSh 4,500 from selling mandazi, and they write a complete, mathematically justified financial plan that accounts for all the decisions Amina must make: allocating her budget across three goals, calculating how much the "20% off" smartphone at the Kondele stall actually costs her, determining what the M-Pesa/mobile-money agent's commission takes from her Uganda transfer, and converting the remaining KES into Uganda Shillings at the correct exchange rate. Every number in the plan must trace back to a formula from a previous lesson, making this a genuine synthesis task rather than a new topic.\n\nThe lesson uses a structured three-stage sequence — individual plan writing, structured peer review using the KICD rubric language, and whole-class share-out — to ensure that every learner produces and critiques a complete financial argument. The peer-review protocol is deliberately modelled on the KICD assessment rubric's four performance levels (Below Expectations, Approaches Expectations, Meets Expectations, Exceeds Expectations), giving learners direct experience of the language used to evaluate their competencies. This metacognitive layer is central to the CBE philosophy: learners are not just answering questions but assessing the quality of mathematical reasoning.\n\nThe lesson closes with two high-visibility rituals that mark the end of the storyline. First, the Driving Question Board receives its final entries: learners write definitive answers — backed by calculations — to the questions posted throughout the unit, and the board moves from a collection of wonderings to a curated record of knowledge built. Second, each learner opens their notebook to the rough explanatory sketch they drew in Lesson 1 and annotates it or redraws it entirely, creating a visual record of conceptual growth. The teacher facilitates a brief gallery-walk comparison so the class can see, literally, how their models have evolved from vague guesses to mathematically precise financial reasoning tools.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Percent word problems: tax and discount Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Arithmetic and Geometric Sequences (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners open their notebooks to their Lesson 1 rough sketch of Amina's financial situation and re-read it silently for 90 seconds. They then individually write a two-sentence prediction: 'If Amina follows all the mathematics we have learned, what will her financial plan look like — and will she have enough money to achieve all three goals?' Learners record their prediction in a dedicated box labelled 'Lesson 8 Starting Prediction' before any group discussion begins.	Project the original Lesson 1 phenomenon text on the board (or read it aloud): 'Amina has KSh 4,500 from selling mandazi. She wants to contribute to the Environmental Club trip to Lake Victoria, buy a second-hand smartphone advertised at 20% off at a Kondele stall, and send the remaining money to her aunt in Uganda.' Say: 'Before we build Amina's final plan, look at the prediction you wrote in Lesson 1 — the very first day. Read it. Think about what you knew then and what you know now.' Allow 90 seconds silent reading. Then say: 'Write your Lesson 8 Starting Prediction in the box on today's page. Do not discuss with your neighbour yet — this is your own thinking.' Allow 2 minutes writing time. After time is up, cold-call 3 learners: 'Zawadi — what is your prediction? Does Amina have enough?' [WAIT TIME: 12 seconds after naming each learner before	Activation of Prior Knowledge via Storyline Reconnection — by returning learners to the identical phenomenon text from Lesson 1, the teacher creates a cognitive 'bookend' that makes the full arc of learning salient. Predictions are recorded in writing so that learners can compare them explicitly with their final plans, making conceptual growth visible and personally meaningful.	Teacher scans written predictions as learners write (circulate for 90 seconds). Look for: (1) learners who reference specific concepts (e.g., 'she will lose some money to commission') — evidence of consolidated understanding; (2) learners who predict only vaguely ('she will have some money left') — flag these learners for closer monitoring during plan writing. Do not correct predictions at this stage; they are diagnostic only.

		accepting the answer.] After each response say: 'Hold that thought — by the end of today you will know whether your prediction was correct, and why.'		
Observe Phase 📹 **VIDEO:** [Percent word problems: tax and discount] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Arithmetic and Geometric Sequences (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners receive the 'Amina's Final Financial Plan' structured task sheet (see teacher notes below for full data set). Working individually for 12 minutes, they complete all five sections of the plan in sequence: (1) Budget — allocate KSh 4,500 across three goals; (2) Discount — calculate the actual sale price of the smartphone after 20% discount; (3) Commission — calculate the mobile-money agent's commission and the net amount transferred to Uganda; (4) Currency Conversion — convert the net KES amount into Uganda Shillings using the printed exchange rate table; (5) Profit/Loss Check — determine whether Amina's mandazi enterprise broke even, made a profit, or made a loss, given her weekly cost price and revenue data. All working must be shown in full. Learners who finish early begin writing a one-paragraph justification explaining which of Amina's three goals is mathematically the riskiest and why.	Distribute the task sheet. Read the scenario data aloud once: 'Here is the data for Amina's plan. Smartphone marked price: KSh 3,200. Discount: 20%. Club trip contribution: KSh 800. Mobile-money agent commission rate: 3% on the amount sent. Exchange rate: KES to UGX — selling rate 28.50 UGX per KES; buying rate 27.80 UGX per KES. Amina's mandazi cost price per week: KSh 1,800 total ingredients and charcoal. Weekly revenue from sales: KSh 2,250. She sold for two weeks to save this money.' Pause and say: 'This is all the information you need. Work through each section in order — do not skip ahead. Show every formula and every substitution, just as we have practised.' [WAIT TIME: Allow learners to begin; circulate silently for the first 3 minutes without intervening.] After 3 minutes, if any learner is stuck on section 1 (budget), prompt quietly: 'What does a budget do? It lists where the money goes — start by writing KSh 4,500 at the top and subtract each goal.' After 8 minutes, say to the whole class: 'You should be finishing section 3 or 4 by now — check your working for section 2 before moving on: did you use Sale Price = Marked Price × (1 − 20/100)?' Cold-call 2 learners to share their section 2 answer only: 'Omondi — what sale price did you get for the smartphone?' [WAIT TIME: 12 seconds.] Confirm: 'KSh 2,560 is correct — KSh 3,200 ×	Sequential Multi-Step Problem Solving with Structured Scaffolding — the task sheet sequences the five sub-tasks in the same order as the unit's lesson sequence (budget → discount → commission → currency → profit/loss), reinforcing the logical dependency chain. Each section requires learners to use the output of the previous section as an input, making the interconnectedness of commercial arithmetic concepts experiential rather than abstract.	During circulation, check three specific indicators: (1) Does the learner's budget in section 1 sum correctly to KSh 4,500? (2) In section 3, does the learner apply commission to the amount being sent (not the full KSh 4,500)? A common error is applying commission to the total savings — flag this immediately with a quiet redirect: 'What amount is Amina actually sending to Uganda? Find that number first.' (3) In section 4, does the learner divide by the selling rate (not the buying rate) when converting KES to UGX? If not, ask: 'Which direction is the money moving — from Kenya to Uganda or Uganda to Kenya? Which rate applies when you are the one buying UGX?'

		0.80 = KSh 2,560. That means the 20% discount saves Amina KSh 640. Now check your budget: does she have KSh 2,560 left after the club contribution?'		
Explain Phase 📹 **VIDEO:** [Percent word problems: tax and discount] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Arithmetic and Geometric Sequences (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners exchange their completed financial plans with a partner (teacher assigns pairs) and conduct a structured peer review using the KICD four-level rubric language. Each reviewer reads the plan carefully and writes specific, evidence-based feedback for each of the five sections using sentence starters provided on the back of the task sheet: 'This section Exceeds/Meets/Approaches/Below Expectations because…' and 'One thing that would improve this section is…'. After 5 minutes of silent review, pairs discuss their feedback face-to-face for 3 minutes, with the plan author listening first before responding. The class then reconvenes for a whole-group share-out in which three pairs present the most significant mathematical disagreement they encountered during peer review.	Say: 'You are now going to do something mathematicians and financial advisors do every day — review someone else's work critically and helpfully. This is not about finding mistakes to embarrass your partner. It is about making both of your plans stronger.' Distribute the peer-review sentence-starter card. Model one example on the board: 'Section 2 — Discount. My partner wrote: Sale Price = 3200 − 20 = 3180. I would say: This section is Below Expectations because the formula used is Discount = Marked Price − 20, which treats the discount as a fixed KSh 20 rather than 20% of the marked price. One thing that would improve this is to use Sale Price = Marked Price × (1 − Percentage Discount ÷ 100), giving Sale Price = 3200 × 0.80 = KSh 2,560.' Say: 'Now read your partner's plan silently and write your feedback. You have 5 minutes — no talking.' [WAIT TIME: enforce silence strictly for 5 minutes; circulate and read feedback as it is written.] After 5 minutes: 'Now face your partner. The plan author stays silent for the first 90 seconds while the reviewer reads their feedback aloud. Then the author responds.' After paired discussion, bring the class back: 'Which pair had the most interesting mathematical disagreement? I want three pairs to share.' Cold-call if no volunteers: 'Fatuma and Akinyi — what did you disagree about?'	Structured Peer Review with Rubric Language — by requiring learners to use the KICD performance-level vocabulary ('Exceeds/Meets/Approaches/Below Expectations') in their feedback, the teacher simultaneously builds metacognitive awareness of quality criteria and familiarises learners with the formal assessment language they will encounter in Grade 10 assessments. The 'author listens first' protocol ensures that feedback is received rather than immediately defended, building the CBE value of Respect.	Teacher reads peer-review feedback cards during the silent review phase, looking for: (1) feedback that is specific and formula-referenced (evidence of deep understanding); (2) feedback that only says 'correct' or 'wrong' without mathematical justification (evidence that the reviewer does not yet understand the criterion — intervene quietly: 'Can you write which formula you expected to see, and whether your partner used it?'); (3) cases where both partners made the same error — these are systemic misconceptions requiring whole-class re-teaching during the share-out.

		[WAIT TIME: 12 seconds.] After each pair shares, ask the class: 'Who else had the same disagreement? Hands up.' Use the frequency of hands to identify the most common conceptual errors and address them directly: 'Three pairs disagreed about which exchange rate to use. Let us settle this now — when Amina sends KES to Uganda, she is purchasing UGX. The agent sells her UGX, so the agent's selling rate applies. She divides KES by 28.50, not 27.80.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Percent word problems: tax and discount] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Arithmetic and Geometric Sequences (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board — the physical display (sticky notes or written cards on a poster) that has accumulated questions, partial answers, and 'I wonder…' statements since Lesson 1. Each learner selects one question from the board that they can now answer definitively using mathematics from their financial plan, writes a final answer card (including at least one formula and one numerical example from Amina's scenario), and posts it next to the original question. The class then does a 3-minute silent gallery walk to read all the final answer cards. Finally, the teacher facilitates a 2-minute whole-class discussion: 'Is there any question on the board that we still cannot fully answer? Are there any new questions that this unit has opened up?'	Direct learners' attention to the DQB. Say: 'Look at this board. In Lesson 1, it was almost empty — just our driving question and a few wonderings. Look at it now.' Allow 20 seconds for learners to observe the board silently. Then say: 'Today we close the board — not because there are no more questions in mathematics, but because we have built enough knowledge to answer the questions we posted. Each of you will write one final answer card. It must include: the question you are answering, the formula you used, and a number from Amina's plan that proves your answer.' Distribute blank cards. Allow 3 minutes for writing. As learners post their cards, circulate and read them. If a card lacks a formula, say quietly: 'Your answer is correct, but show me the mathematics — which formula gives this result?' After all cards are posted, say: 'Three minutes — silent gallery walk. Read every answer card. If you see an error, put a small question mark sticker on it — we will discuss those.' After the gallery walk, ask: 'Did anyone place a	Driving Question Board Closure with Evidence-Based Final Answers — the DQB closure ritual makes the cumulative architecture of the unit's storyline visible and permanent. Requiring learners to post formula-backed answer cards transforms the board from a collection of questions into a structured knowledge artefact, reinforcing the CBE principle that learning outcomes must be demonstrated through evidence, not just stated.	Teacher reads all final answer cards posted on the board, categorising them mentally into three groups: (1) cards with correct formula and correct numerical answer — Meets or Exceeds Expectations; (2) cards with correct formula but arithmetic error — Approaches Expectations; flag the learner's name for written feedback on their task sheet; (3) cards with no formula or an incorrect formula — Below Expectations; speak individually to these learners during the Model Building phase. Question-mark stickers placed by peers during the gallery walk provide a second layer of peer formative assessment data.

		question mark? Let us hear why.' [WAIT TIME: 12 seconds.] Address flagged cards briefly. Then ask: 'Here is a harder question — does any card on this board make you think of a new question we have NOT answered yet in Grade 10?' Cold-call 2 learners. [WAIT TIME: 12 seconds each.] Likely responses: 'What happens if the exchange rate changes every day?'; 'How do insurance companies make profit?' Affirm: 'Excellent — those are questions for Grade 11 and beyond. You have just shown that real mathematical thinking never stops.'		
Model Building Phase 📹 **VIDEO:** [Percent word problems: tax and discount] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Arithmetic and Geometric Sequences (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners open their notebooks to the rough explanatory model or diagram they drew in Lesson 1 — their initial attempt to map Amina's financial decisions. Using a different colour pen, they now annotate or completely redraw the model to show: (a) each financial operation (budget, discount, commission, profit/loss, currency conversion) labelled with its correct formula; (b) the flow of Amina's KSh 4,500 through each decision, with correct amounts at each stage; (c) a final box showing the outcome for each of Amina's three goals — Environmental Club trip contribution: KSh 800 ✓; Smartphone purchase: KSh 2,560 (possible, but leaves only KSh 1,140 for the Uganda transfer); Uganda transfer: KSh 1,140 − commission = net amount converted to UGX. Learners write one final sentence at the bottom of their model: 'The answer to our driving question is…' This sentence must name at least three of the six operations covered and	Say: 'In Lesson 1, you drew a rough map of Amina's problem. You did not know the formulas yet — you were guessing. Now you know everything. Open to that first diagram.' Allow 30 seconds for learners to locate it. Then say: 'Using your red or green pen — whatever colour you have that is different from your original — annotate your Lesson 1 model or draw a new one beside it. Show me: the formula for each operation, the correct amounts at each stage, and the final outcome for each of Amina's three goals. You have 5 minutes.' [WAIT TIME: allow silent individual work for the first 3 minutes; circulate and observe without intervening unless a learner is completely stuck.] After 3 minutes, say: 'You should be adding the flow of money — how does KSh 4,500 shrink or transform as it moves through each decision?' After 5 minutes, ask learners to write their final driving-question answer sentence. Cold-call 3 learners to read their	Cumulative Model Revision with Annotated Comparison — by requiring learners to annotate their Lesson 1 model in a contrasting colour rather than replace it, the teacher makes conceptual growth literally visible on the page. The side-by-side comparison activates metacognition (learning to learn) and fulfils the CBE requirement for learners to reflect on their own development. The final driving-question answer sentence consolidates synthesis and prepares learners for formal written assessment.	Collect all task sheets and final annotated models at the end of the lesson. Use the KICD four-level rubric to evaluate each learner's financial plan against all five criteria (budget, discount, commission, currency conversion, profit/loss). Look for: (1) complete, correctly sequenced plans with all formulas shown — Meets or Exceeds Expectations; (2) plans where one or two sections are missing formulas or contain carried-forward errors from an earlier section — Approaches Expectations; (3) plans where three or more sections are incomplete — Below Expectations; these learners require targeted remediation before the sub-strand assessment. Note also the quality of the final driving-question answer sentence: a high-quality sentence names all six operations and explains how they interact; a weak sentence names operations without explaining their relationship to Amina's specific decisions.

	explain how they collectively enable or constrain Amina's decisions.	final sentence aloud: 'Cherotich — read me your final answer to the driving question.' [WAIT TIME: 12 seconds.] After each reading, ask the class: 'Does this answer include at least three operations? Give a thumbs-up if yes.' Use thumbs to identify incomplete answers and prompt: 'What is missing? Which operation did she not mention?' Close the lesson by saying: 'Compare your Lesson 1 model and your Lesson 8 model side by side. What is the biggest difference?' Cold-call 2 learners. [WAIT TIME: 12 seconds each.] End with: 'The biggest difference is not just more labels — it is that now your model is backed by mathematics. That is what commercial arithmetic gives you: the power to make smart, fair, and informed money decisions. That is what Amina needed, and that is what you now have.'		

D. TEACHER REFLECTION
1. When learners completed Amina's five-section financial plan, which of the six operations (budget, discount, commission, profit/loss, currency conversion) generated the most errors? Does this suggest that a particular lesson (1–7) needs to be redesigned or extended in future cycles? 2. During peer review, did learners use the KICD rubric language ('Exceeds/Meets/Approaches/Below Expectations') meaningfully, or did they default to vague praise and criticism? What explicit modelling would help future cohorts give more precise, formula-referenced feedback? 3. How many learners' Lesson 8 final models showed a qualitative improvement over their Lesson 1 sketches? Were there learners whose models became more detailed but less accurate — adding labels without correct values — and what does this reveal about their conceptual understanding versus procedural fluency? 4. Did the Driving Question Board closure reveal any questions that the unit had not adequately addressed? If learners posted answer cards with persistent errors (e.g., wrong exchange rate direction, commission applied to the wrong base), which lessons in the sequence should be revised to address those specific misconceptions earlier? 5. Reflecting on the full eight-lesson arc: did the Amina storyline maintain genuine explanatory relevance throughout, or did it become a superficial wrapper around decontextualised calculations in some lessons? How might the phenomenon be refined — perhaps by introducing a second character (e.g., Amina's aunt in Uganda responding to the transfer) — to sustain deeper engagement through all eight lessons in future delivery?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed their own mathematical growth by comparing their rough Lesson 1 model of Amina's financial problem — drawn

	before any instruction — with their fully annotated Lesson 8 model, which correctly traces KSh 4,500 through budget allocation, a 20% discount on a KSh 3,200 smartphone (sale price KSh 2,560), a 3% mobile-money commission on the Uganda transfer, and a final currency conversion into Uganda Shillings at the correct selling rate of 28.50 UGX/KES. They also observed peers' financial plans and identified errors or omissions using the KICD rubric language.
What did I learn?	Learners consolidated all six commercial arithmetic learning outcomes — budget preparation, discount and percentage discount, commission and percentage commission, percentage profit and loss, and currency conversion — into a single integrated financial plan. They learned that these operations are not independent: the amount available for currency conversion depends on the budget allocation, which in turn depends on the correct discount calculation and commission deduction. They also learned that smart financial decision-making requires checking each step before proceeding to the next, and that the direction of a currency transaction (KES → UGX vs. UGX → KES) determines which exchange rate applies.
How does this explain the phenomenon?	Amina's puzzle — how the same KSh 4,500 produces such different outcomes depending on discounts, commissions, exchange rates, and profit/loss — is now fully explained: each operation applies a specific mathematical formula to a specific base amount, and small percentage differences (e.g., using a buying rate of 27.80 instead of a selling rate of 28.50) produce real financial gains or losses. Amina — and any Kenyan — can make smart, fair, and informed money decisions by applying these six formulas correctly, in the right sequence, to the right base values, at every stage of a financial transaction.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.